

TECHNOLOGICAL INSTITUTE OF THE PHILIPPINES - MANILA

College of Computer Studies

**Enhancing Thematic Literary Analysis of Rizal's Novels Through
Hybrid Information Retrieval: Integrating Lexical and Semantic Approaches
Using XLM-RoBERTa Embeddings for Noli Me Tangere and El Filibusterismo**

Bachelor of Science in Computer Science

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TABLE OF CONTENTS

CHAPTER 1 INTRODUCTION	1
1.1 Background of the Study	1
1. Subjectivity and inconsistency in manual thematic analysis	10
3. Lack of specialized computational literary studies tools for Filipino texts.....	11
1.3 Research Questions	12
1.4 Objectives of the Study.....	12
General Objective	12
Specific Objectives.....	13
1.5 Significance of the Study	13
1.5.6 Application Utilization and Impact Metrics	15
1.6 Scope and Delimitation.....	22
1.7 Definition of terms.....	25
1.8 Basis of the Study.....	27
CHAPTER 2 REVIEW OF RELATED LITERATURE AND STUDIES	33
2.1 Hybrid Retrieval Models: Lexical–Semantic Integration.....	33
2.1.1 Justification for Hybrid Architecture in Filipino Literary Retrieval	34
2.2 Redundancy Management in Multi-Document Retrieval.....	41
2.3 Preprocessing and Stopword Management in Multilingual Contexts	41
2.4 Multilingual Semantic Models: XLM-RoBERTa and Cross-Lingual Understanding.....	42
2.5 Domain-Adaptive Pre-training for Specialized Tasks.....	45
2.6 Scoring Functions and Ranking Mechanisms	46
2.7 Thematic Retrieval and Topic Modeling.....	46
2.8 The Research Gap	48
2.9 Theoretical Framework.....	50
2.9.1 Hybrid Retrieval Theory	50
2.9.2. Cross-lingual Transfer Learning Theory.....	51
2.9.3 Thematic Coherence Theory.....	51
2.9.4 Integration of Theoretical Frameworks.....	52
2.10 Synthesis of the Study	52
2.10.1 Key Findings and Convergence	52

2.10.2 Research Gaps Addressed by This Study	53
2.10.3 Contribution to the Field	55
2.10.4 Positioning Within the Research Landscape	55
CHAPTER 3 METHODOLOGY	57
3.1 Research Design	57
3.3 Conceptual Framework.....	60
3.3.1 Input Layer: Query and Corpus.....	62
3.3.2 Processing Layer: Domain-Adaptive Semantic Validation	62
3.3.3 Processing Layer: Dual-Formula Hybrid Retrieval Architecture	65
3.3.4 Analysis Layer: Thematic Classification and Context Expansion.....	67
3.3.5 Output Layer: Retrieval Results and Visualization	68
3.3.6 Theoretical Integration	70
3.2.7 Framework Validation	71
3.4. Textual Corpus.....	72
3.4.1 Rationale for Using Summary Versions	72
3.4.2 Thematic Data Sources.....	73
3.4.3 Domain-Adaptive Corpus Representation.....	74
3.5 Text Cleaning and Normalization	77
3.5.1 Tagalog Stopword Management.....	77
3.5.2 Tokenization for XLM-RoBERTa.....	80
3.5.3 Passage Segmentation Strategy.....	80
3.5.4 Deduplication	82
3.5.5 Context Expansion with Neighbor Formula.....	82
3.6 System Architecture.....	84
3.6.1 Overall Architecture.....	84
3.6.2 Query Validation Component	86
3.6.3 Multi-Sentence Query Validation Protocol.....	89
3.6.4 Semantic Retrieval Component	91
3.6.5 Lexical Retrieval Component	96
3.6.6 Dual-Formula Hybrid Integration	98
3.6.7 Formula Application Example	101
3.6.8 Query Suggestion Component.....	105
3.6.8.1 Single-Word Theme Input	105

3.6.8.2 Single-Word Character Input.....	107
3.6.8.3 Entity-Based Chapter Location with Domain-Adaptive Semantic Grounding.....	108
Entity Detection and Validation.....	109
Chapter Mapping Process.....	109
Contextual Information.....	109
Display Format.....	110
Application Examples.....	110
System Integration.....	113
3.8 Evaluation Methodology.....	116
3.8.1 Baseline Comparison.....	117
3.8.2 Quantitative Evaluation Metrics.....	118
3.8.3 Qualitative Expert Review.....	120
3.8.4 Thematic Distribution and Co-occurrence Analysis.....	121
3.8.5 Query Complexity Testing.....	122
Simple Character-Action Queries (Basic Relational Patterns).....	124
Multi-Concept & Multi-Relation Queries (Complex Semantic Patterns).....	124
3.9 Implementation Details.....	126
3.9.1 Development Environment and Tools.....	126
3.9.2 Data Storage Format.....	127
3.9.3 System Workflow.....	127
3.9.4 System Parameters.....	133
3.10 Ethical Considerations.....	134
3.11 Limitations and Delimitations.....	136
3.11.1 Methodological Limitations.....	136
3.11.2 Delimitations.....	137
3.11.3 Validation Boundaries.....	139
3.12 Summary.....	140
References.....	144

List of Figures

Figure 1: Theoretical Framework of a Hybrid Document Retrieval System Leveraging Semantic and Lexical Matching Based on Kuzi et al. (2020).....	50
Figure 5: Data Collection and Preprocessing Pipeline.....	59
Figure 2: Conceptual Framework for the Hybrid Information Retrieval System Using XLM-RoBERTa for Thematic Literary Analysis of Rizal's Novels	61
Figure 3: Domain-Adaptive Pre-training (DAPT) Validation Architecture for Semantic Grounding ..	64
Figure 4: Dataset Preparation and Annotation Process.....	76
Figure 6: System Architecture of the Hybrid Information Retrieval System Using XLM-RoBERTa for Rizal's Novels	84
Figure 7: Hybrid Retrieval System Architecture (Detailed Component View).....	86
Figure 8: Model Training and Fine-Tuning Process.....	93
Figure 9: Sentence-Level Embedding Generation.....	95
Figure 10: Thematic Classification and Analysis Process.....	114
Figure 11: Evaluation Metrics and Framework	117
Figure 12: System Testing and Validation Workflow.....	123
Figure 13: Query Processing Execution Workflow.....	129

List of Tables

Table 1: Mga Tanong sa Survey para sa Pangangailangan ng Information Retrieval System (N=100)	17
Table 2: Traditional vs. Hybrid Information Retrieval Systems	40
Table 3: Word-by-Word Weight Assignment (Example 1)	78
Table 4: Word-by-Word Weight Assignment (Example 2)	79

CHAPTER 1

INTRODUCTION

1.1 Background of the Study

The literary works of José Rizal, particularly *Noli Me Tangere* (1887) and *El Filibusterismo* (1891), are foundational texts in Philippine national identity and continue to serve as essential components of Filipino education and cultural discourse (Rizal, 1887, 1891). These novels present complex thematic structures addressing colonialism, social justice, education, religion, and revolution, interwoven through intricate narratives that reflect the socio-political landscape of Spanish colonial Philippines. The depth and complexity of these themes have sustained scholarly interest for over a century, yet traditional literary analysis methods remain predominantly manual, subjective, and limited in their capacity to systematically identify and compare thematic patterns across the extensive text of both novels.

Understanding the computational challenges this study addresses requires establishing the historical context of José Rizal's foundational works and the educational reality faced by Filipino students. *Noli Me Tangere* (1887) follows Crisostomo Ibarra, a young Filipino reformist who returns from Europe with idealistic dreams of building a school in San Diego, symbolizing education as national progress. Key characters include Maria Clara (Ibarra's beloved representing Filipino women trapped between tradition and modernity), Padre Damaso (corrupt Franciscan friar symbolizing clerical abuse), Elias (revolutionary advocating violent resistance), and Sisa (tragic mother driven to madness by systemic injustice). The novel critiques Spanish colonial oppression through interconnected themes of *edukasyon* (education as both hope and threat), *relihiyon* (institutional

corruption versus genuine faith), kolonyalismo (colonial control pervading Filipino life), katarungan (justice and legal injustice), and pamilya at dangal (family honor preserving identity while enabling colonial control).

El Filibusterismo (1891) serves as a darker sequel, tracing idealism's transformation into revolutionary fervor. Simoun, revealed as the disguised embittered Ibarra, now plots violent overthrow of colonial government as a wealthy jeweler. Basilio, Sisa's surviving son and now a medical student, must choose between peaceful reform and revolutionary violence. The novel explores rebolusyon vs. reporma (revolution versus reform as central debate), karahasan (violence examining revolutionary violence's moral legitimacy), kasakiman (greed showing how colonial exploitation corrupts both colonizers and colonized), pagkabigo (disillusionment representing idealism's death under sustained oppression), and kamatayan at sakripisyo (death and sacrifice questioning whether martyrdom serves revolution or wastes lives). The two novels function as complementary halves of Rizal's social critique—*Noli Me Tangere* diagnoses colonial society's diseases while *El Filibusterismo* explores potential cures and their costs.

The narrative and thematic complexity of these works creates specific challenges for literary analysis and education. Thematic evolution means the same theme manifests differently across novels and within narrative progression—"edukasyon" shifts from Ibarra's initial optimism to later disillusionment, requiring contextual understanding that distinguishes these variations. Character transformation requires distinguishing Ibarra's characterization in *Noli Me Tangere* from his Simoun identity in *El Filibusterismo*. Multi-layered symbolism carries meaning beyond literal narrative—Sisa's madness represents not just personal tragedy but systemic violence against Filipino families. Intertextual references cross-reference between novels, requiring recognition when *El Filibusterismo* passages reference *Noli Me Tangere* events.

In contemporary Filipino education, students studying these mandatory texts face persistent difficulties navigating their extensive length and complex thematic structures. Teachers report that many students struggle to complete full chapter readings, often relying on online summaries that miss nuanced character development and thematic interconnections. When assigned to identify passages supporting specific themes such as "edukasyon bilang pag-asa" (education as hope) or "karahasan laban sa reporma" (violence versus reform), students encounter difficulty navigating texts 15,000-20,000 words, resulting in responses lacking textual evidence or complete task abandonment. The vocabulary mismatch problem becomes evident in student search behaviors: searching "edukasyon" misses semantically related passages discussing "pag-aaral" (studying), "kaalaman" (knowledge), or "karunungan" (wisdom). Character confusion persists—questions about character fates and relationships appear repeatedly in student forums, indicating students lack systematic methods to verify plot details against textual sources.

Contemporary literary analysis faces significant methodological challenges in the digital age. While humanities scholars have access to increasingly sophisticated computational tools, the application of these technologies to Filipino literary texts remains underdeveloped, particularly for low-resource languages like Tagalog (Cosme & De Leon, 2024). Computational linguistics has rarely been applied to Filipino literary studies, despite substantial advances in natural language processing (NLP) for other languages. This gap becomes particularly evident when examining thematic analysis, where computational methods could potentially reveal patterns, connections, and thematic distributions that manual close-reading approaches might overlook or fail to systematically quantify.

The traditional system of analyzing *Noli Me Tangere* and *El Filibusterismo* relied heavily on manual close reading methods. Researchers would physically read through both novels, manually identifying themes, taking handwritten or typed notes, and organizing their findings using paper-

based systems or basic word processing tools. This approach was time-consuming and limited in scope, as scholars could only process information at the speed of human reading and manual categorization. The identification of thematic patterns required extensive memory work and cross-referencing between different sections of the texts, often leading to potential oversights of subtle connections that appeared across large textual distances.

In contrast, the hybrid system proposed in this study combines traditional literary analysis with modern computational methods. This approach maintains the interpretive depth of human scholarship while leveraging technology to handle large-scale pattern detection and data organization. The hybrid system uses advanced natural language processing to automatically identify and categorize thematic elements across both novels, significantly reducing the time required for initial thematic mapping. However, unlike purely automated systems, this hybrid approach preserves human oversight at critical stages. Researchers validate the machine-identified themes, provide contextual interpretation, and make final judgments about thematic significance and relationships. This integration allows scholars to focus their expertise on interpretation and analysis rather than on mechanical tasks of text searching and pattern counting, ultimately producing more comprehensive and nuanced insights into Rizal's works.

The evolution of information retrieval systems has demonstrated that hybrid approaches combining lexical exact-match methods with semantic understanding yield superior performance compared to single-method approaches (Gao et al., 2021; Kuzi et al., 2020). Lexical retrieval models such as BM25 excel at capturing explicit keyword correspondences but fail to recognize semantic relationships when different terminology expresses similar concepts. Conversely, neural embedding models capture semantic similarity but may overlook exact matches that carry specific significance in literary contexts. The integration of these complementary approaches has shown particular

promise in multilingual contexts, where linguistic diversity compounds the challenges of information retrieval (Wiciaputra et al., 2021).

Recent advances in cross-lingual transformer models, particularly XLM-RoBERTa (Cross-lingual Language Model - Robustly Optimized BERT Approach), have demonstrated remarkable capabilities in handling multilingual and code-switched text (Salve & Tubil, 2025; Timoneda & Vallejo Vera, 2023). XLM-RoBERTa's pre-training on 100 languages enables transfer learning from high-resource languages to low-resource languages like Filipino, creating shared semantic spaces where conceptually related content clusters together regardless of language (Conneau et al., 2019). Imperial (2021) demonstrated that BERT-based embeddings could effectively substitute for language-specific NLP tools in Filipino contexts where such tools remain unavailable or underdeveloped, achieving performance improvements of up to 12.4% in F1 scores compared to classical approaches.

The application of topic modeling and thematic retrieval systems has shown transformative potential in religious and literary text analysis. Ignaco and Ballera (2025) demonstrated that traditional keyword-based search achieved 0% accuracy in identifying thematically related Bible verses when exact keywords were absent, while their Latent Dirichlet Allocation (LDA)-based approach achieved 33-67% accuracy by capturing topical relationships beyond surface-level lexical matching. More sophisticated approaches integrating transformer-based embeddings with topic modeling have shown even greater promise, with Aamir et al. (2025) demonstrating that BERTopic combined with XLM-R outperformed traditional topic modeling methods for Urdu text, a low-resource language with linguistic challenges comparable to Filipino.

The linguistic complexity of Rizal's novels presents unique challenges for computational analysis. Written primarily in Tagalog with significant Spanish influence and, in contemporary educational contexts, often studied alongside English translations, these texts exemplify the code-switching phenomenon that characterizes much Filipino discourse (Cosme & De Leon, 2024). Effective computational analysis must therefore accommodate multilingual elements, historical language variation, and the cultural embeddedness of thematic content. Furthermore, the novels' thematic richness—spanning colonialism (*kolonyalismo*), oppression (*pang-aapi*), education (*edukasyon*), revolution (*rebolusyon*), religion (*relihiyon*), social justice (*katarungan*), and family honor (*pamilya at dangal*)—requires analytical approaches capable of capturing both explicit thematic markers and implicit conceptual relationships.

Existing computational literary studies have predominantly focused on Western canonical texts in high-resource languages, particularly English (Gaikwad et al., 2024). The limited application of advanced NLP techniques to Filipino literature reflects broader patterns of digital humanities research, where low-resource languages remain underserved despite possessing rich literary traditions worthy of computational investigation. This disparity extends beyond mere technological application; it represents a gap in cultural representation within digital humanities scholarship, where Filipino literary heritage has yet to benefit from the analytical tools increasingly applied to Western texts.

The development of hybrid information retrieval systems specifically designed for Filipino literary texts addresses multiple interconnected needs. For educators and students, such systems could enhance comprehension and analysis of mandatory literary texts by enabling thematic exploration beyond traditional chapter-by-chapter reading. For scholars, computational tools could facilitate comparative thematic analysis, revealing patterns across Rizal's works and enabling more

systematic investigation of thematic evolution, distribution, and interconnection. For cultural researchers and historians, computational literary analysis offers methods to quantitatively examine how specific historical and cultural themes manifest throughout canonical texts, providing empirical foundations for interpretive claims.

The technical foundation for this research rests on several converging developments in NLP and information retrieval. XLM-RoBERTa has demonstrated state-of-the-art performance across diverse Filipino text processing tasks, achieving 97.62% precision, 95.77% F1-score, and 97.50% ROC-AUC in code-switched Tagalog-English classification when properly fine-tuned (Salve & Tubil, 2025). The model's architecture enables capture of contextual semantics and cross-lingual relationships essential for thematic analysis. Simultaneously, advances in hybrid retrieval architectures have established frameworks for integrating semantic embeddings with lexical matching and language-specific preprocessing (Gao et al., 2021). The availability of Filipino-specific stopwords lists through resources like stopwords-iso-tl addresses a critical preprocessing requirement identified as essential for optimal performance in Tagalog text processing (Bation et al., 2017).

The significance of developing computational literary analysis tools for Filipino texts extends beyond immediate research applications. Such tools contribute to the broader project of digital humanities inclusivity, ensuring that advanced analytical methods serve diverse linguistic and cultural contexts rather than reinforcing existing biases toward high-resource languages. Moreover, by demonstrating effective approaches for low-resource literary text analysis, this research provides methodological foundations that can be adapted to other underserved languages and literary traditions throughout Southeast Asia and globally.

The thematic structure of Rizal's novels, while extensively studied through traditional literary criticism, has not been subjected to systematic computational analysis that could reveal quantitative patterns in thematic distribution, co-occurrence, and textual manifestation. Themes such as edukasyon (education) appear throughout both novels but manifest differently in various narrative contexts, character dialogues, and plot developments. Computational thematic analysis could map these distributions, identify passages where multiple themes converge, and enable comparison of how themes evolve between *Noli Me Tangere* and *El Filibusterismo*. Such capabilities would provide educators and students with powerful tools for exploring these complex texts while offering scholars new methodologies for literary analysis grounded in quantitative evidence.

Research Gap

The thematic structure of Rizal's novels, while extensively studied through traditional literary criticism, has not been subjected to systematic computational analysis that could reveal quantitative patterns in thematic distribution, co-occurrence, and textual manifestation. Themes such as edukasyon (education) appear throughout both novels but manifest differently in various narrative contexts, character dialogues, and plot developments. **Despite advances in hybrid information retrieval systems and multilingual NLP models, no computational system has been developed to enable sophisticated thematic exploration of these foundational Filipino literary texts.** This absence represents a critical gap in both Filipino digital humanities and computational literary studies, limiting the analytical tools available to researchers, educators, and students engaging with these culturally significant works. Specifically:

1. **No passage-level semantic retrieval exists for Filipino literary works**, despite extensive research on general documents in high-resource languages

2. **Hybrid retrieval systems have not been adapted to low-resource languages** like Filipino with limited NLP tools
3. **Thematic retrieval using Filipino-capable embeddings remains unexplored**, leaving educators without computational support for theme-based analysis
4. **Filipino-specific preprocessing pipelines** (stopword removal, code-switching handling, cultural terminology) have not been systematically integrated into literary retrieval systems
5. **Domain-adaptive pre-training for Filipino literature** has not been validated, despite proven effectiveness in other specialized domains

Addressing these gaps requires developing a hybrid information retrieval system that integrates semantic understanding through multilingual models with Filipino-specific linguistic preprocessing, enabling systematic, reproducible thematic analysis that complements rather than replaces traditional literary scholarship.

1.2 Statement of the Problem

The analysis of thematic content in José Rizal's *Noli Me Tangere* and *El Filibusterismo* currently relies predominantly on manual close-reading methods and basic keyword-based searching (Ctrl+F) that, while valuable for deep interpretive work, suffer from inherent limitations in systematicity, scalability, and reproducibility. Traditional keyword searching can only locate exact word matches and fails to capture semantic relationships, synonyms, or contextual meanings essential for comprehensive thematic analysis. Despite the availability of advanced natural language processing technologies and the proven effectiveness of hybrid information retrieval systems in other linguistic contexts, no computational system has been developed to enable sophisticated thematic

exploration of these foundational Filipino literary texts. This absence represents a critical gap in both Filipino digital humanities and computational literary studies, limiting the analytical tools available to researchers, educators, and students engaging with these culturally significant works.

Specifically, the following problems characterize the current state of thematic literary analysis for Rizal's novels:

1. Subjectivity and inconsistency in manual thematic analysis

Traditional literary analysis of *Noli Me Tangere* and *El Filibusterismo* depends on individual readers' interpretation and memory to identify thematically relevant passages throughout the texts. This manual approach introduces subjective variation in which passages are deemed relevant to particular themes, making systematic comparison across different analytical perspectives difficult. The extensive length of both novels (approximately 15,000-20,000 words combined) exacerbates this challenge, as readers cannot feasibly maintain comprehensive awareness of all thematically relevant passages throughout both texts simultaneously. Furthermore, the lack of computational tools means that claims about thematic prevalence, distribution, or co-occurrence rest on qualitative impressions rather than quantifiable evidence, limiting the empirical foundation of literary scholarship on these works.

2. Inability to capture semantic thematic relationships beyond explicit keyword matching

Current approaches to locating thematic content in Rizal's novels rely primarily on searching for explicit keywords or manually reading through entire texts to identify relevant passages. However, themes manifest through diverse linguistic expressions, metaphors, narrative descriptions, and dialogues that may not contain the exact thematic keywords. For example, passages addressing

kolonyalismo (colonialism) may discuss specific manifestations of colonial oppression without using the term "colonialism" itself. Similarly, themes of katarungan (justice) may appear in narrative descriptions of injustice, legal proceedings, or moral deliberations using varied vocabulary. The absence of semantic retrieval capabilities means that thematically relevant passages are systematically missed when they do not contain expected keywords, resulting in incomplete thematic analysis that fails to capture the full richness and complexity of Rizal's thematic treatment.

3. Lack of specialized computational literary studies tools for Filipino texts

While computational literary analysis has become increasingly sophisticated for English and other high-resource languages, Filipino literature remains severely underserved by digital humanities tools and methodologies (Cosme & De Leon, 2024; Imperial, 2021). Existing information retrieval systems are not designed to handle the linguistic characteristics of Filipino literary texts, including historical language variation and culturally embedded terminology. General-purpose search tools fail to incorporate Filipino-specific linguistic preprocessing such as Tagalog stopword removal, which has been shown to improve text processing performance by 35-45% in corpus size reduction while enhancing accuracy (Ladani & Desai, 2020; Bation et al., 2017). This technological gap perpetuates the marginalization of Filipino literature within computational literary studies and deprives Filipino students, educators, and scholars of analytical tools comparable to those available for Western canonical texts.

These interconnected problems create a situation where one of the Philippines' most culturally significant literary corpus remains largely inaccessible to modern computational analysis methods, despite the demonstrated effectiveness of such methods in other linguistic and literary

contexts. The absence of specialized tools limits both the depth of scholarly analysis possible and the pedagogical resources available for teaching these mandatory texts in Philippine education.

1.3 Research Questions

This study addresses the following research questions:

1. How can a hybrid information retrieval system combining semantic embeddings and lexical matching address the subjectivity and inconsistency inherent in manual thematic analysis of *Noli Me Tangere* and *El Filibusterismo*?
2. To what extent can XLM-RoBERTa semantic embeddings capture thematic relationships beyond explicit keyword matching in identifying relevant passages across Rizal's novels?
3. How does integrating Filipino-specific linguistic preprocessing with XLM-RoBERTa's cross-lingual capabilities improve retrieval accuracy for Filipino literary texts compared to generic preprocessing approaches?

1.4 Objectives of the Study

General Objective

To develop and evaluate a hybrid information retrieval system integrating XLM-RoBERTa semantic embeddings with lexical matching and Filipino-specific linguistic preprocessing for enhanced thematic literary analysis of José Rizal's *Noli Me Tangere* and *El Filibusterismo*.

Specific Objectives

1. To design and implement a hybrid retrieval system that provides consistent, quantifiable thematic analysis of *Noli Me Tangere* and *El Filibusterismo*, reducing subjective variation through computational scoring mechanisms.
2. To evaluate the effectiveness of XLM-RoBERTa semantic embeddings in identifying thematically relevant passages that lack explicit keywords, comparing semantic retrieval performance against traditional keyword-based approaches.
3. To integrate Tagalog-specific stopwords filtering and cross-lingual transfer learning capabilities into a specialized computational tool for Filipino literary text analysis, demonstrating improved retrieval accuracy for culturally embedded terminology.

1.5 Significance of the Study

The development of a hybrid information retrieval system for thematic analysis of Rizal's novels provides valuable benefits across multiple stakeholder groups in Philippine education, scholarship, and cultural preservation.

For Literary Scholars and Researchers

This study introduces a computational methodology for systematic thematic analysis of Filipino literature. The hybrid system enables efficient retrieval of theme-relevant passages across both *Noli Me Tangere* and *El Filibusterismo*, facilitating comparative analysis that would be impractical manually. By identifying thematic co-occurrence patterns, scholars can investigate the complexity of Rizal's social commentary and interconnections within his colonial critique. The

quantitative evidence supports literary arguments with systematic data rather than selective examples, and this methodology can extend to other works in the Filipino literary canon.

For High School Students

Filipino students mandated to study Rizal's novels gain an accessible exploration tool for these complex texts. The system helps students efficiently locate passages relevant to specific themes like edukasyon (education) or pang-aapi (oppression), enhancing comprehension and engagement. This targeted retrieval capability particularly supports essay writing and exam preparation, making lengthy texts more navigable and reducing frustration while improving learning outcomes.

For Historians and Cultural Researchers

The computational analysis provides quantitative insights into historical and cultural theme representation in foundational Philippine texts. Mapping thematic distribution reveals attention patterns, thematic evolution between novels, and theme conjunctions that illuminate Rizal's conceptualization of colonial society. This offers evidence for claims about priorities in Filipino nationalist discourse during the late Spanish colonial period and demonstrates digital humanities applications for Philippine cultural heritage materials.

For Educators in Literature and History

The system serves as a didactic tool for enhanced classroom instruction. Educators can efficiently gather thematically organized passages for discussion and close reading, use visualizations to make abstract concepts concrete, and facilitate independent student investigations

that promote active learning and critical thinking. This integration demonstrates digital literacy relevance while maintaining focus on literary understanding and cultural heritage.

For Future Computational Linguistics Researchers

This study provides methodological contributions extending beyond Rizal's novels. It validates that XLM-RoBERTa embeddings effectively apply to Filipino literary texts despite limited language-specific training data, supporting cross-lingual transfer learning for low-resource contexts (Imperial, 2021; Salve & Tubil, 2025). The hybrid architecture establishes a framework adaptable to other low-resource languages and literary traditions. The evaluation methodology offers a model for rigorous assessment where ground truth is subjective, and demonstrates that advanced NLP techniques can serve culturally diverse contexts when appropriately adapted, challenging the field's historical bias toward high-resource languages (Cosme & De Leon, 2024; Asai et al., 2022).

Broader Cultural Impact

The study contributes to cultural preservation and digital heritage accessibility in the Philippines. By developing computational tools for Filipino literary analysis, it ensures Philippine cultural heritage remains accessible and relevant digitally. This methodology can extend to digitized collections of Philippine literature, historical documents, and cultural texts, enabling sophisticated search and analysis capabilities that preserve Filipino cultural heritage while increasing accessibility for both Filipino and international audiences.

1.5.6 Application Utilization and Impact Metrics

The practical significance of this research is evidenced through a comprehensive needs assessment survey conducted at Carlos Botong Francisco Memorial National Highschool during the

planning phase of system development. This assessment aimed to validate the necessity of a hybrid information retrieval system for Filipino literary texts and to understand student challenges in studying Rizal's novels.

Survey Methodology and Participant Demographics

The survey was administered to 100 students (50 Grade 9, 50 Grade 10) enrolled in Filipino subjects at Carlos Botong Francisco Memorial National Highschool, Angono Rizal, during November 2025. The participant group comprised 58 female students and 42 male students, ages 14-16, representing diverse socioeconomic backgrounds typical of Philippine public secondary education. All participants had been studying *Noli Me Tangere* and *El Filibusterismo* as part of mandatory curriculum requirements under Republic Act 1425 (Rizal Law). The survey employed a 5-point Likert scale where 1 = *Lubos na Hindi Sumasang-ayon* (Strongly Disagree), 2 = *Hindi Sumasang-ayon* (Disagree), 3 = *Walang Kinikilingan* (Neutral), 4 = *Sumasang-ayon* (Agree), and 5 = *Lubos na Sumasang-ayon* (Strongly Agree). All 100 students completed the survey during designated class time, achieving 100% response rate.

Quantitative Survey Results

The survey results demonstrate overwhelming student recognition of the need for computational support in literary analysis, with 99% of students expressing agreement that an information retrieval system would significantly benefit their study of Rizal's novels. Detailed results for each survey item appear in Table 1.

Table 1: Mga Tanong sa Survey para sa Pangangailangan ng Information Retrieval System (N=100)

Tanong sa Survey	Lubos na di Sumasang-ayon	Hindi Sumasang-ayon	Walang Kinikilingan	Sumasang-ayon	Lubos na Sumasangayon	Mean	SD
	(1)	(2)	(3)	(4)	(5)		
1. Makakatulong sa akin ang information retrieval system upang mabilis na makahanap ng mga talata tungkol sa mga tema tulad ng edukasyon, katarungan, at kolonyalismo.	0 (0%)	0 (0%)	2 (2%)	28 (28%)	70 (70%)	4.68	0.49
2. Mas magiging interesado ako sa pag-aaral ng Noli at El Fili kung may teknolohiya akong magagamit para sa paghahanap ng mga character at tema.	0 (0%)	1 (1%)	6 (6%)	35 (35%)	58 (58%)	4.49	0.68
3. Kailangan ko ng tulong ng sistema para masubaybayan ang mga tauhan mula sa Noli Me Tangere hanggang El Filibusterismo.	0 (0%)	0 (0%)	4 (4%)	38 (38%)	58 (58%)	4.54	0.58
4. Makakatipid ako ng oras sa paggawa ng essays at reading journals kung may search system para sa mga nobela ni Rizal.	0 (0%)	0 (0%)	3 (3%)	32 (32%)	65 (65%)	4.62	0.54
5. Gusto kong malaman agad kung aling kabanata ang may kaugnayan sa tema na aking hinahanap nang hindi na kailangang basahin ang buong nobela.	0 (0%)	1 (1%)	5 (5%)	41 (41%)	53 (53%)	4.46	0.65

Overall System Need Assessment: When responses were aggregated, 99 students (99%) rated their need for such a system as 4 or 5 (*kailangan o lubos na kailangan*), with an overall mean rating of 4.56 ($SD = 0.59$). Only one student (1%) provided a neutral rating (3), with zero students expressing disagreement regarding system utility.

Analysis of Survey Findings

The survey results reveal consistent patterns across multiple dimensions of student need. The highest perceived benefit (mean = 4.68) emerged for fast thematic search capability, with 70% of students strongly agreeing that the system would help them quickly locate passages about themes such as education, justice, and colonialism. This finding directly validates the vocabulary mismatch problem identified in the problem statement, where students struggle to find thematically relevant passages using current manual or keyword-only methods.

Time-saving value received strong endorsement (mean = 4.62), with 65% strongly agreeing that the system would save time in completing essays and reading journals. This response aligns with teacher reports that students often abandon analytical tasks perceived as too time-consuming when passage location requires manual navigation through 15,000-20,000 words combined pages. The time efficiency benefit represents a critical factor for students managing heavy academic workloads across multiple subjects within the constraints of Philippine public-school schedules.

Character tracking need (mean = 4.54) indicates that 58% of students strongly agree they require systematic support to follow characters across both novels. This validates documented confusion about character arcs, relationships, and transformations—exemplified by the persistent question "Namatay ba si Basilio?" appearing in student forums despite Basilio's survival being a

crucial plot element. The character tracking capability addresses not merely convenience but a fundamental comprehension barrier preventing students from understanding narrative continuity between *Noli Me Tangere* and *El Filibusterismo*.

Increased motivation through technology (mean = 4.49) demonstrates that 58% of students strongly agree their interest in Filipino literature would increase with access to modern retrieval tools. This finding has important pedagogical implications: technology integration serves not only functional purposes (finding passages efficiently) but also motivational purposes (increasing engagement with culturally significant texts). The 6% neutral responses on this item, higher than other questions, suggest individual variation in how students perceive technology's role in literary study—a diversity the system must accommodate through optional rather than mandatory use.

Direct chapter relevance identification (mean = 4.46) shows that 53% strongly agree they want to immediately know which chapters relate to search themes without reading entire novels. This response reflects a pragmatic student approach to literary analysis: rather than comprehensive linear reading (which fewer than 30% complete according to teacher reports), students seek efficient pathways to locate information relevant to specific analytical tasks. The system's ranking mechanisms, which identify chapters with highest thematic relevance, directly address this expressed need.

Demographic and Accessibility Considerations

Survey analysis revealed no statistically significant differences in perceived need across gender (female mean: 4.58, male mean: 4.53, $p > 0.05$) or grade level (Grade 9 mean: 4.55, Grade 10 mean: 4.57, $p > 0.05$), suggesting that computational support addresses universal rather than demographically specific challenges. Informal follow-up questions regarding technology access

confirmed that approximately 95% of surveyed students own smartphones with internet capability, validating the mobile-first design approach. Students from households without broadband internet (estimated 40-45% based on socioeconomic indicators) expressed equivalent interest in the system, provided it functions efficiently on mobile data networks—a requirement incorporated into system design through network optimization and efficient query processing.

Qualitative Feedback from Open-Ended Responses

The survey included an optional open-ended question: "Ano ang pinakamahirap sa pag-aaral ng Noli at El Fili?" (What is most difficult about studying Noli and El Fili?). Sixty-three students provided google form responses, revealing consistent themes:

Time and Length Challenges (mentioned by 42 students, 67%): "Sobrang haba ng nobela, hindi ko matapos basahin lahat." (The novels are too long, I can't finish reading everything.) "Walang oras para basahin ang buong chapter bago klase." (No time to read whole chapters before class.)

Theme Identification Difficulties (mentioned by 28 students, 44%): "Hindi ko alam saan makikita yung mga tema sa bawat kabanata." (I don't know where to find the themes in each chapter.) "Mahirap hanapin yung example ng katarungan o edukasyon sa nobela." (Hard to find examples of justice or education in the novels.)

Character Confusion (mentioned by 31 students, 49%): "Nakakalito kung sino-sino yung mga tauhan, marami silang pangalan." (Confusing who the characters are, they have many names.) "Hindi ko alam kung ano nangyari kay Basilio sa dalawang nobela." (I don't know what happened to Basilio in both novels.)

Evidence Location Problems (mentioned by 24 students, 38%): "Hirap maghanap ng mga quote para sa essay." (Hard to find quotes for essays.) "Kailangan ng textual evidence pero di ko alam saan hahanapin." (Need textual evidence but don't know where to find it.)

These qualitative responses corroborate the quantitative findings and provide rich contextual understanding of student challenges that computational retrieval systems can address. The frequency with which students mention time constraints, difficulty locating themes, and challenges finding textual evidence validates the research problem statement and establishes clear user requirements for system design.

Synthesis of Needs Assessment

The comprehensive survey data encompassing Likert-scale quantitative responses, demographic analysis, and qualitative open-ended feedback establishes a strong empirical foundation for the necessity of developing a hybrid information retrieval system for Filipino literary texts. The 99% agreement rate regarding system utility, combined with mean scores consistently above 4.4 across all measured dimensions, demonstrates that student need for computational support is not marginal but overwhelming. The alignment between quantitative ratings and qualitative explanations strengthens confidence that the survey captured genuine student experiences rather than mere acquiescence to researcher expectations.

These findings validate that the challenges identified in the problem statement subjectivity and inconsistency in manual analysis, inability to capture semantic relationships beyond keywords, and lack of specialized tools for Filipino texts—reflect authentic educational difficulties experienced by the target user population. The survey results provide empirical justification for the research investment required to develop, implement, and evaluate the proposed hybrid retrieval system,

establishing that such a system addresses documented needs rather than offering solutions seeking problems.

Furthermore, the near-universal smartphone ownership and expressed willingness to use mobile-accessible literary tools confirm the technological feasibility of deployment within Philippine educational contexts. The system's anticipated utilization is supported not merely by researcher assumptions but by direct evidence from 100 students representing the target user demographic, providing confidence that the completed system will achieve meaningful adoption and educational impact when deployed at scale.

1.6 Scope and Delimitation

This study focuses specifically on the development and evaluation of a hybrid information retrieval system for thematic analysis of two novels by José Rizal: *Noli Me Tangere* (1887) and *El Filibusterismo* (1891). The scope encompasses the following elements:

Textual Corpus

The study utilizes Filipino language summary versions (buod) of both novels from publicly available educational resources. Specifically, the *Noli Me Tangere* summary from PinoyCollection.com (2017) and the *El Filibusterismo* summary from Noypi.com.ph (2019). These summarized versions represent accessible educational materials commonly used by Filipino students and provide manageable text lengths for system development and testing.

Thematic Focus

The research examines multiple major themes central to Rizal's novels, identified from established literary analysis sources. Themes for *Noli Me Tangere* were derived from GradeSaver (2024) and Francisco (2025), while themes for *El Filibusterismo* were sourced from LitCharts (2025) and GradeSaver (2023). These themes were selected based on their prominence in established literary scholarship and their significance in Philippine cultural and historical discourse.

Technical Scope

The system architecture integrates XLM-RoBERTa multilingual embeddings (specifically the xlm-roberta-base model) for semantic representation, combined with lexical keyword matching and Tagalog-specific stopwords filtering using the stopwords-iso-tl resource. The retrieval process operates at the passage level, with passages defined as semantically coherent text segments of variable length. The scoring mechanism incorporates semantic similarity (via cosine similarity of XLM-RoBERTa embeddings), lexical overlap (via keyword presence detection), and length-based normalization to balance relevance against passage length.

Evaluation Methodology

System performance is evaluated through retrieval accuracy assessment across multiple themes, comparing the hybrid approach against baseline keyword-only retrieval. Evaluation includes analysis of precision and recall, examination of thematic distribution patterns, and qualitative assessment of retrieved passage relevance through expert review. The study also includes comparative analysis of how different scoring components contribute to overall retrieval effectiveness.

Delimitations

Use of Summary Versions

The study does not utilize the complete, full-length versions of *Noli Me Tangere* and *El Filibusterismo*. This limitation is due to the original texts containing mixed Spanish-Tagalog content and their considerable length, which would complicate processing and analysis. Instead, the research focuses on Filipino language summaries that provide comprehensive coverage of major plot points and themes while maintaining linguistic consistency and manageable scope.

Model Training Limitations

The study does not include automated fine-tuning of the XLM-RoBERTa model on Filipino literary texts due to the limited availability of annotated Filipino literature datasets suitable for such fine-tuning. Instead, the research uses the pre-trained model's cross-lingual transfer capabilities, following approaches validated by Imperial (2021) and Salve and Tubil (2025) for Filipino text processing.

Scope of Literary Coverage

The system does not perform automated theme identification or discovery; rather, it retrieves passages relevant to predefined themes specified by users. The research does not extend to other works by Rizal or other Filipino authors, maintaining focus on the two novels most central to Philippine literary education.

System Functionality Boundaries

The system does not provide literary interpretation or critical analysis of retrieved passages; it serves as a retrieval and exploration tool to support rather than replace human literary scholarship. The evaluation does not include large-scale user studies with students or educators, focusing instead on technical performance assessment and expert validation. Future work may extend the methodology to additional Filipino literary texts and conduct comprehensive user studies to assess effectiveness in educational contexts.

1.7 Definition of terms

The following terms are defined as they are used in this study:

Cosine Similarity: A metric measuring the similarity between two vectors by calculating the cosine of the angle between them, ranging from -1 to 1, commonly used to assess semantic similarity between text embeddings.

Cross-lingual Transfer Learning: The application of knowledge learned from high-resource languages to low-resource languages through shared multilingual representations, enabling effective NLP for languages with limited training data (Wiciaputra et al., 2021; Imperial, 2021).

Digital Humanities: An interdisciplinary field applying computational methods and digital technologies to humanities research, including literary studies, history, and cultural analysis (Gaikwad et al., 2024).

Filipino Literature: Literary works written in Filipino (standardized Tagalog) or Philippine languages, encompassing novels, poetry, essays, and other forms, with particular focus on works contributing to Philippine national identity and cultural heritage.

Hybrid Information Retrieval: An approach combining multiple retrieval methodologies, specifically integrating semantic embedding-based retrieval with lexical keyword matching to use complementary strengths of both approaches (Gao et al., 2021; Kuzi et al., 2020).

Lexical Matching: Text retrieval based on exact or approximate matching of keywords and terms, identifying passages that contain specified words or phrases regardless of semantic interpretation.

Low-resource Language: A language with limited availability of digital corpora, annotated datasets, and NLP tools compared to high-resource languages like English, presenting challenges for computational linguistic applications (Asai et al., 2022; Cosme & De Leon, 2024).

Multi-component Scoring: A ranking approach that combines multiple relevance signals—such as semantic similarity, lexical overlap, and length normalization—into a unified score for ranking retrieved passages (Yoo et al., 2024).

Passage Retrieval: The task of identifying and ranking text segments (passages) from a larger corpus based on their relevance to a query or theme, operating at a granularity between sentence-level and document-level retrieval.

Semantic Embeddings: Dense vector representations of text that encode semantic meaning, enabling computation of semantic similarity between texts based on conceptual relatedness rather than lexical overlap (Imperial, 2021; Gaikwad et al., 2024).

Stopwords: Common words in a language that carry minimal semantic content and are typically removed during text preprocessing to improve retrieval performance, such as "ang," "mga," "si," and "dahil" in Tagalog (Bation et al., 2017; Ladani & Desai, 2020).

TF-IDF (Term Frequency–Inverse Document Frequency): A numerical statistic reflecting the importance of a word in a document relative to a corpus, used to weight terms in information retrieval and text mining (Bation et al., 2017; AlShammari, 2023).

Thematic Analysis: The identification and interpretation of recurring themes—abstract concepts, topics, or ideas—throughout literary texts, examining how themes manifest, develop, and interrelate across narratives (Ignaco & Ballera, 2025; Aamir et al., 2025).

Thematic Distribution: The pattern of how a theme appears throughout a text or corpus, including frequency, concentration in specific sections, and co-occurrence with other themes.

XLM-RoBERTa (Cross-lingual Language Model – Robustly Optimized BERT Approach): A multilingual transformer-based language model pre-trained on 100 languages that enables cross-lingual transfer learning and generates contextual embeddings for text in multiple languages including Filipino (Conneau et al., 2019; Salve & Tubil, 2025).

1.8 Basis of the Study

This research is grounded in multiple intersecting foundations that establish both its necessity and feasibility: empirical evidence from Philippine educational contexts, technological advances in multilingual natural language processing, educational mandates codified in Philippine law, theoretical frameworks from information retrieval science, and cultural equity considerations within digital humanities scholarship.

1.8.1 Empirical Basis: Documented Limitations in Filipino Literary Education

The study's foundation rests on empirically documented challenges in Philippine secondary education. Teachers report that fewer than 30% of students complete assigned chapter readings, with the majority relying on online summaries that omit crucial narrative details and thematic nuances. When assignments require locating passages supporting specific themes such as "edukasyon bilang pag-asa" or "karahasan laban sa reporma," students struggle to navigate texts exceeding 15,000-20,000 words combined pages, resulting in vague responses lacking textual evidence or complete task abandonment. The vocabulary mismatch problem becomes evident through student search behaviors: searching "edukasyon" fails to retrieve content discussing "pag-aaral," "kaalaman," or "karunungan"—all thematically relevant but lexically distinct. Character tracking difficulties reveal persistent confusion about character relationships and transformations, with questions about character fates appearing repeatedly in student forums despite plot elements being clearly established in the texts. These documented educational challenges reflect broader patterns in Philippine literary education where students lack systematic methods to verify plot details or locate thematic content efficiently.

1.8.2 Technological Basis: Advances in Multilingual NLP

The study's feasibility rests on recent breakthroughs in cross-lingual language modeling that enable sophisticated NLP for low-resource languages like Filipino. Multiple independent studies establish XLM-RoBERTa as optimal for Filipino text processing. Imperial (2021) demonstrated 12.4% F1-score improvement over classical approaches on Filipino readability assessment, establishing that BERT embeddings substitute effectively for unavailable language-specific NLP tools. Cosme and De Leon (2024) achieved 0.84 accuracy on Filipino-English code-switched sentiment analysis,

dramatically outperforming lexicon-based monolingual tools. Salve and Tubil (2025) attained 97.62% precision and 95.77% F1-score on Taglish classification, demonstrating XLM-RoBERTa's superior handling of bilingual Filipino text. These findings establish that XLM-RoBERTa's pre-training on 100 languages creates shared semantic spaces enabling effective Filipino processing through cross-lingual transfer learning. The integration of semantic and lexical approaches through hybrid retrieval is validated by Gao et al. (2021), who demonstrated that semantic embeddings trained to capture what lexical models miss achieve superior performance through complementary integration, and Kuzi et al. (2020), who showed that hybrid models significantly improve recall during retrieval and precision during reranking. Krieger et al. (2022) established that pre-trained models benefit substantially from domain-specific adaptation, with performance gains proportional to domain alignment between pre-training and target tasks, suggesting XLM-RoBERTa's general capabilities can be enhanced through semantic grounding in Rizal's novels specifically.

1.8.3 Educational Basis: Mandated Curriculum Requirements

The study addresses concrete educational needs arising from Philippine curriculum mandates. Republic Act 1425 (Rizal Law), enacted in 1956, legally mandates that courses on José Rizal's life, works, and writings be included in curricula of all Philippine schools, colleges, and universities. Section 1 explicitly requires that *Noli Me Tangere* and *El Filibusterismo* be included in public and private education. At the secondary level, the K-12 curriculum integrates Rizal's novels into Filipino literature courses, requiring students to demonstrate abilities to analyze literary texts for theme development, compare thematic treatments across works, evaluate how historical contexts shape literature, and construct evidence-based interpretations supported by textual citations. Teachers and students report persistent difficulties including text length intimidation with combined 400+ pages overwhelming students managing heavy academic workloads; thematic complexity with

interconnected themes challenging synthesis; limited computational support with no existing tools helping students efficiently locate and compare passages; and assessment difficulties when essay questions require textual evidence across both novels. These educational imperatives establish practical urgency for computational tools that reduce barriers to engaging with culturally foundational texts.

1.8.4 Theoretical Basis: Information Retrieval Frameworks

The study's architecture rests on established theoretical frameworks from information retrieval science. Hybrid Retrieval Theory (Gao et al., 2021; Kuzi et al., 2020) proposes that optimal retrieval emerges from integrating complementary lexical and semantic approaches rather than treating them as competing alternatives. Cross-lingual Transfer Learning Theory (Conneau et al., 2019) explains how multilingual transformer models develop language-agnostic representations that transfer effectively across typologically diverse languages, enabling sophisticated NLP for low-resource languages. Thematic Coherence Theory (Yoo et al., 2024; Ignaco & Ballera, 2025) establishes that document relevance operates at multiple levels—lexical, semantic, and thematic—each contributing distinct information requiring explicit modeling and appropriate weighting. Domain-Adaptive Pre-training Theory (Krieger et al., 2022) demonstrates that pre-trained models benefit from domain-specific adaptation when pre-training and target domains share structural similarities, improving task performance while preventing false retrievals from semantically misaligned queries. These frameworks collectively guide the dual-formula hybrid architecture, XLM-RoBERTa model selection, multi-component scoring mechanisms, and domain-adaptive validation thresholds implemented in this study.

1.8.5 Cultural Basis: Digital Humanities Inclusivity

The study addresses fundamental disparity in digital humanities where Filipino literature remains severely underserved by computational analysis tools despite the Philippines' rich literary tradition. Computational literary studies demonstrate overwhelming concentration on English-language texts and European traditions, with over 95% of published research focusing on high-resource languages (Gaikwad et al., 2024). Filipino lacks basic NLP tools such as part-of-speech taggers, dependency parsers, and named entity recognizers available for English. This technological gap perpetuates cultural marginalization within digital scholarship. Developing computational tools for Filipino literature ensures Philippine literary heritage remains accessible in the digital age, provides methodological foundations adaptable to other Southeast Asian literary traditions, and challenges the field's historical bias toward Western canonical texts. The study contributes to broader digital humanities inclusivity by demonstrating that advanced analytical methods can serve diverse linguistic and cultural contexts rather than reinforcing existing biases toward high-resource languages.

1.8.6 Synthesis: Convergent Foundations

These multiple bases converge to establish both necessity and feasibility. Necessity emerges from empirical evidence documenting severe limitations in current approaches, RA 1425 legal mandates creating curriculum requirements, pedagogical standards demanding evidence-based analysis, and cultural equity considerations calling for computational tools serving diverse literatures. Feasibility emerges from XLM-RoBERTa's demonstrated effectiveness for Filipino text processing, validated hybrid retrieval architectures providing proven frameworks, theoretical foundations guiding principled system design, and domain-adaptive pre-training enabling semantic

grounding in specific literary works. The intersection of these foundations establishes that this study addresses a genuine research gap at an opportune technological moment, with clear practical applications and broader implications for digital humanities inclusivity. This multi-layered basis justifies the research investment and positions the study within larger conversations about language equity in computational scholarship, technology's role in education, and the future of literary studies in digital contexts.

CHAPTER 2

REVIEW OF RELATED LITERATURE AND STUDIES

The development of effective information retrieval systems requires understanding both traditional keyword-based methods and modern semantic approaches to text processing, particularly for multilingual and low-resource language contexts. This review examines the progression of research from basic information retrieval models to specialized applications in Filipino/Tagalog text processing, establishing a theoretical and empirical foundation for the present study.

2.1 Hybrid Retrieval Models: Lexical–Semantic Integration

The evolution of information retrieval systems has been marked by a gradual shift from purely keyword-based retrieval to hybrid approaches that incorporate semantic understanding. Traditional keyword-based retrieval systems, while computationally efficient, have long suffered from vocabulary mismatch problems and inability to capture semantic relationships between queries and documents.

Gao et al. (2021) addressed this fundamental limitation through the development of CLEAR (Complement Lexical Retrieval Model with Semantic Residual Embeddings), which trains neural embeddings to capture language structures and semantics that lexical retrieval fails to identify. Their residual-based embedding learning method demonstrated substantial improvements over classical BM25 models, achieving enhanced accuracy and efficiency in reranking pipelines. This work established that semantic and lexical signals are not competing but complementary dimensions of information retrieval.

Building upon this foundation, Kuzi et al. (2020) expanded understanding of hybrid retrieval systems by examining their application across the two-phase retrieval paradigm—initial document retrieval followed by reranking. Their empirical study using TREC collections revealed that combining deep neural network-based semantic models with keyword matching-based lexical models significantly improved recall rates during the retrieval stage. Critically, they demonstrated that semantic approaches excel at capturing conceptual similarities while lexical approaches maintain precision for exact-match requirements. This complementary nature justified the integration of both approaches in modern retrieval architectures, particularly for complex query scenarios where semantic understanding becomes essential.

The theoretical foundation for semantic-lexical integration was further strengthened by Melamud et al. (2015), who developed a word embedding model for lexical substitution that used context embeddings from the skip-gram architecture. Their work demonstrated that explicit use of context embeddings—previously considered merely internal components of the learning process—could achieve state-of-the-art results in identifying meaning-preserving word substitutes. This finding had important implications for information retrieval, suggesting that embeddings capture detailed semantic relationships that extend beyond simple word co-occurrence patterns.

2.1.1 Justification for Hybrid Architecture in Filipino Literary Retrieval

The necessity of a hybrid approach combining semantic embeddings with lexical matching becomes evident when examining the fundamental limitations of single-method retrieval systems in Filipino literary contexts.

The Inadequacy of Pure Lexical Approaches

Traditional keyword-based retrieval systems operate on exact string matching, identifying passages containing specified terms regardless of semantic context. This approach encounters three critical problems. First, vocabulary mismatch occurs because themes manifest through diverse expressions without exact keywords—a passage discussing "kolonyalismo" might describe forced labor or cultural suppression without using "kolonya" itself. Second, synonym blindness treats "katarungan," "patas na paghatol," and "hustisya" as distinct terms despite their semantic equivalence. Third, context insensitivity cannot distinguish "edukasyon" as Ibarra's hope versus the friars' fear—identical words carrying opposite thematic meanings.

The Limitations of Pure Semantic Approaches

Relying solely on neural embeddings introduces different deficiencies. Semantic models may retrieve conceptually related passages lacking specific terminology users expect, reducing result trust. This computational opacity undermines pedagogical objectives when students cannot understand why passages without explicit keywords were retrieved. Additionally, semantic systems risk false associations based on spurious correlations rather than genuine thematic relationships.

The Complementary Strengths of Hybrid Integration

The hybrid approach leverages both methodologies' strengths. XLM-RoBERTa embeddings capture thematic meaning and conceptual relationships beyond keywords, while weighted term matching ensures explicit terminology receives appropriate prioritization. The dual-formula architecture validates character-action relationships against canonical events, preventing absurd results like "kabaitan ni Padre Damaso" for consistently antagonistic characters. For complex queries

such as "pang-aapi ng mga prayle sa pamilya ni Ibarra," spaCy dependency parsing verifies that actions, agents, and targets semantically align in contextually correct relationships.

Computational Trade-offs and Their Justification

This architectural sophistication introduces computational costs: embedding generation, similarity computation, multi-component scoring, and domain validation. However, these costs are justified when result quality outweighs speed. Hybrid retrieval achieves significantly higher precision and recall than keyword-only baselines. Students benefit more from semantically appropriate results than immediate but incomplete matches requiring manual filtering. The system remains deployable on standard educational hardware without GPU requirements.

As Gao et al. (2021) and Kuzi et al. (2020) demonstrated, performance gains from hybrid integration justify computational overhead when retrieval quality impacts downstream tasks. This principle applies more strongly to educational applications in low-resource languages, where retrieval errors compound learning difficulties. When students struggle to locate relevant passages, they may abandon sophisticated analysis for superficial observations a pedagogical failure outweighing computational costs.

This foundation establishes hybrid architecture as a fundamental requirement for effective thematic literary analysis in Filipino texts. The integration of semantic understanding with lexical precision, validated through domain-adaptive pre-training and multi-component scoring, supports complex analytical tasks while maintaining computational feasibility in educational environments.

Comparison of Traditional and Hybrid Information Retrieval Systems

Information retrieval in traditional literary analysis operates through lexical matching processes where documents are retrieved based on exact term correspondence between queries and indexed content. The Boolean model, which represents the classical approach to information retrieval, treats documents as sets of terms and employs logical operators (AND, OR, NOT) to specify retrieval conditions (Sciencedirect.com, n.d.). Traditional methods such as BM25 (Best Matching 25) and TF-IDF (Term Frequency-Inverse Document Frequency) calculate relevance scores based on statistical measures of term frequency, inverse document frequency, and document length normalization (Iterate.ai, n.d.; Mtham8.github.io, n.d.). While these approaches excel at exact keyword matching with query processing times under 10 milliseconds, they face fundamental limitations in semantic understanding (Google Cloud, 2024). The vocabulary mismatch problem—where semantically relevant documents use different terminology than the search query—represents the primary weakness of traditional systems (University of South Africa, 2021). For instance, a search for "cardiovascular disease" would fail to retrieve documents containing the synonymous term "heart condition," despite their semantic equivalence.

In contrast, hybrid information retrieval systems integrate lexical precision with semantic understanding through multiple parallel retrieval pathways. These systems combine sparse vector indices (using BM25 or learned sparse models like SPLADE) for keyword matching with dense vector indices (using transformer-based embeddings from BERT or Sentence Transformers) for semantic retrieval (Zilliz.com, 2024; Meilisearch, 2023). The architecture employs neural encoding pipelines that represent documents in both high-dimensional dense vectors (768-1024 dimensions) and sparse vectors with thousands of dimensions but few non-zero values (Google Cloud, 2024; Zilliz.com, 2024). A fusion layer, typically implementing Reciprocal Rank Fusion (RRF), merges

result from both retrieval methods based on ranking positions rather than raw scores, avoiding the complexities of score normalization across different algorithms (Microsoft Learn, 2024; OpenSearch, 2024). This dual approach addresses the vocabulary mismatch problem by understanding context and meaning beyond exact term matches, enabling the system to recognize synonyms, related concepts, and contextual variations automatically (Whisperit.ai, n.d.).

Empirical evidence demonstrates substantial performance advantages for hybrid systems across standardized benchmarks. Research by Siriwardhana et al. (2024) on the Natural Questions (NQ) dataset showed hybrid approaches achieved an NDCG@10 score of 0.67 compared to 0.633 for baseline methods, representing a 5.8% improvement, while the TREC-COVID dataset demonstrated even greater gains with NDCG@10 of 0.87 versus 0.804 baseline, an 8.2% improvement. On the SQuAD dataset, hybrid retrieval achieved an F1 score of 68.4 compared to 52.63 for fine-tuned baseline systems, marking a 30% improvement (Siriwardhana et al., 2024). Comparative studies report that semantic search components improve retrieval precision by 25-35% over keyword methods for ambiguous queries and reduce irrelevant results by up to 40% in enterprise knowledge bases (Whisperit.ai, n.d.). Industry implementations have shown response time reductions of 30-40% and retrieval accuracy increases of 40% in production environments (Dev.to, 2024). Rakuten's implementation of semantic search reported a 5% increase in sales, demonstrating tangible business impact beyond academic metrics (Celerdata.com, n.d.).

However, hybrid systems introduce significant computational and implementation complexities that must be considered. Dense vector indices consume substantially more memory—approximately 50GB for 5 million documents in the HotPotQA dataset—compared to the minimal storage requirements of traditional inverted indices (Siriwardhana et al., 2024). Query processing

latency increases to 10-100 milliseconds depending on the fusion method employed, compared to sub-10 millisecond response times for traditional systems (OpenSearch, 2024; Google Cloud, 2024). The implementation requires careful tuning of fusion parameters, weight optimization between keyword and semantic components, and often necessitates GPU resources for acceptable performance with neural embedding models (Fuzzylabs.ai, n.d.; Couchbase, 2024). Performance dependencies on embedding model quality mean that general-purpose models may underperform in specialized domains without fine-tuning, and bi-encoder architectures offer minimal advantages over BM25 in zero-shot, out-of-domain scenarios (University of Waterloo, 2024; Emergentmind.com, n.d.). Storage overhead for advanced architectures like ColBERT can reach 6-10 times larger footprints than traditional indices (arXiv, 2021).

The choice between traditional and hybrid retrieval systems depends on specific application requirements and resource constraints. Traditional systems remain appropriate for scenarios requiring exact term matching, minimal computational overhead, and situations where vocabulary is standardized and controlled. Hybrid systems demonstrate clear advantages for applications involving natural language queries, diverse terminology, cross-lingual searches, and contexts where semantic understanding is critical—such as literary analysis of texts with rich figurative language, historical vocabulary variations, and complex thematic interconnections. For analyzing works like *Noli Me Tangere* and *El Filibusterismo*, where themes manifest through various narrative devices, symbolic imagery, and evolving terminology across the two novels, hybrid retrieval's capacity to identify semantically related passages regardless of exact lexical matches provides substantial analytical advantages that justify the increased computational investment

Table 2: Traditional vs. Hybrid Information Retrieval Systems

Aspect	Traditional IR	Hybrid IR	Sources
Core Method	Lexical matching with Boolean operators, BM25, TF-IDF	Lexical matching + semantic understanding via neural embeddings	Sciencedirect.com (n.d.); Meilisearch (2023)
Index Structure	Inverted index (sparse)	Multiple indices: inverted + dense vector	Google Cloud (2024); Zilliz.com (2024)
Query Processing Time	<10ms	10-100ms	OpenSearch (2024); Google Cloud (2024)
Storage Requirements	Minimal (inverted index only)	Significant (50GB for 5M documents)	Siriwardhana et al. (2024)
Semantic Understanding	None—exact term matching only	Yes—context, synonyms, related concepts	Whisperit.ai (n.d.)
Vocabulary Mismatch	Cannot resolve	Effectively addresses	University of South Africa (2021)
NDCG@10 Performance (NQ)	0.633	0.67 (+5.8%)	Siriwardhana et al. (2024)
NDCG@10 Performance (TREC-COVID)	0.804	0.87 (+8.2%)	Siriwardhana et al. (2024)
F1 Score (SQuAD)	52.63	68.4 (+30%)	Siriwardhana et al. (2024)
Precision Improvement	Baseline	+25-35% for ambiguous queries	Whisperit.ai (n.d.)
Implementation Complexity	Low	High—requires parameter tuning, fusion optimization	Fuzzylabs.ai (n.d.); Couchbase (2024)
Hardware Requirements	CPU sufficient	Often requires GPU for embeddings	Couchbase (2024)
Best Use Cases	Exact matching, controlled vocabulary	Natural language, diverse terminology, semantic search	Dev.to (2024)

2.2 Redundancy Management in Multi-Document Retrieval

As retrieval systems evolved to handle larger document collections, the challenge of redundancy detection became increasingly critical. Hendrickx et al. (2009) investigated this problem in the context of multi-document summarization for Dutch, introducing semantic overlap detection tools that went beyond simple string matching. While their initial results did not demonstrate superiority over traditional methods, their work highlighted a fundamental tension in information retrieval: the need to balance comprehensive coverage with the elimination of redundant information. This challenge becomes particularly acute in passage-based retrieval systems where multiple segments may convey similar information through different word choices.

2.3 Preprocessing and Stopword Management in Multilingual Contexts

The effectiveness of any retrieval system fundamentally depends on appropriate text preprocessing, particularly regarding stopwords handling and term extraction. Ladani and Desai (2020) provided a comprehensive survey of stopwords identification and removal techniques, documenting that stopwords removal can decrease corpus size by 35-45% while improving efficiency and accuracy of text mining applications. Their analysis revealed that most stopwords resources have been developed for high-resource languages, with limited attention to language-specific stopwords lists for low-resource languages. This gap poses significant challenges for multilingual retrieval systems.

Kalykulova and Alzhanov (2025) advanced the field by introducing UNI-TEX, an unsupervised method for unigram term extraction that uses embedding-based filtering to address

data noise caused by common words. Testing on the ACTER corpus across four domains and three languages demonstrated an average F1-score of 46% for English, establishing the viability of semantic filtering for term extraction. However, their work also highlighted the persistent challenge of adapting such methods to languages with limited NLP resources and tools.

For Tagalog specifically, Bation et al. (2017) demonstrated the practical importance of stopwords removal in document classification tasks. Their automatic categorization system for Tagalog documents using Support Vector Machines achieved 92% accuracy when stopwords such as "ang," "mga," "si," and "dahil" were removed prior to feature extraction. The study established that these high-frequency Tagalog words, while grammatically essential, offer minimal discriminative power for document categorization—a finding with direct implications for information retrieval systems operating on Filipino text.

The role of word frequency in retrieval quality was further examined by AlShammari (2023), who implemented text similarity measurement using word frequency and cosine similarity in Python. Their work demonstrated that term weighting based on frequency distributions provides a computationally efficient mechanism for relevance scoring. However, the English-centric nature of most frequency-based approaches necessitates validation on other language corpora, particularly for languages like Tagalog where frequency distributions may exhibit different characteristics.

2.4 Multilingual Semantic Models: XLM-RoBERTa and Cross-Lingual Understanding

The advent of multilingual transformer models represented a paradigm shift in cross-lingual information retrieval, with XLM-RoBERTa emerging as a particularly powerful architecture for low-resource languages. Conneau et al. (2019) developed XLM-RoBERTa as a cross-lingual language

model trained on 100 languages, enabling transfer learning from high-resource languages to low-resource languages through shared semantic representations.

Wiciaputra et al. (2021) explored XLM-RoBERTa's transfer learning capabilities for bilingual text classification in English and Indonesian, achieving 93.3% accuracy on English datasets and 90.2% on Indonesian datasets through a mixed-language training approach. Their results demonstrated that XLM-RoBERTa's cross-lingual representations effectively captured semantic patterns across related languages, suggesting potential applicability to Filipino/Tagalog contexts.

Imperial (2021) provided crucial evidence for XLM-RoBERTa's effectiveness in Filipino NLP tasks through research on automatic readability assessment. By combining BERT embeddings with handcrafted linguistic features, Imperial achieved a 12.4% improvement in F1 performance over classical approaches on Filipino datasets. Significantly, the study demonstrated that BERT embeddings could serve as substitute feature sets for low-resource languages with limited semantic and syntactic NLP tools—a finding directly relevant to the challenges of Filipino information retrieval where traditional linguistic resources are scarce.

The applicability of XLM-RoBERTa to Filipino and Tagalog was further validated by Asai et al. (2022) in their cross-lingual open-retrieval question answering shared tasks spanning 16 typologically diverse languages, including Tagalog. While the best-performing system achieved 32.2 F1 overall, performance on Tagalog remained challenging, with some systems yielding near-zero scores. This highlighted the persistent difficulties of information retrieval in truly low-resource contexts and underscored the need for specialized approaches tailored to Filipino linguistic characteristics.

Cosme and De Leon (2024) provided definitive evidence for XLM-RoBERTa's superiority in handling Filipino-English code-switching through their sentiment analysis study of product and service reviews. Their fine-tuned XLM-RoBERTa model achieved 0.84 accuracy and weighted F1-score, with individual class F1-scores of 0.89, 0.86, and 0.78 for positive, negative, and neutral sentiments respectively. The dramatic performance gap between XLM-RoBERTa and lexicon-based tools designed for monolingual text showed the limitations of single-language approaches when applied to the bilingual reality of Filipino communication. This finding has direct implications for retrieval systems targeting Filipino literature, which frequently exhibits code-switching patterns.

The performance advantages of XLM-RoBERTa over alternative transformer architectures were systematically documented by Salve and Tubil (2025) in their study of bilingual code-switching. Testing XLM-RoBERTa Large against both its Base variant and multilingual BERT (mBERT) on Taglish text classification tasks, they achieved 97.62% precision, 95.77% F1-score, and 97.50% ROC-AUC. The study's advanced preprocessing techniques—including language tagging, ambiguity tagging, and customized tokenization—enabled the model to handle informal and code-switched constructs effectively. These results established XLM-RoBERTa Large as particularly well-suited for Filipino text processing applications.

Comparative analyses across multiple transformer architectures by Timoneda and Vallejo Vera (2023) provided broader context for model selection in political science text analysis. Their cross-lingual evaluation demonstrated that XLM-RoBERTa significantly outperformed both multilingual BERT and multilingual DeBERTa in cross-lingual applications, though RoBERTa and DeBERTa showed advantages over BERT in certain monolingual circumstances. This reinforced XLM-RoBERTa's position as the optimal choice for multilingual retrieval tasks involving low-resource languages.

2.5 Domain-Adaptive Pre-training for Specialized Tasks

The enhancement of pre-trained language models through domain-specific adaptation has emerged as a critical strategy for improving performance in specialized applications. Krieger et al. (2022) explored domain-adaptive pre-training (DAPT) for language bias detection in news, demonstrating how equipping pre-trained models with domain-specific knowledge significantly improves task performance. Their approach built upon foundational work by Sun et al. and Gururangan et al., which established that adapting BERT and RoBERTa to domains related to the target task substantially boosts classification accuracy, while irrelevant domain pre-training leads to performance decline.

Krieger et al. (2022) performed domain-adaptive pre-training on the WNC corpus, consisting of 180,000 manually annotated sentence pairs contrasting biased and neutral statements from Wikipedia. The corpus included epistemological, framing, and demographic bias types. Given that both WNC (used for pre-training) and BABE (used for fine-tuning) shared similar bias structures, their hypothesis posited that domain-specific pre-training would substantially enhance model performance in sentence-level media bias detection. Their focused and direct setup leveraged large-scale bias-related data to achieve improved accuracy and robustness in bias classification, demonstrating that domain adaptation strategies previously successful in biomedical (BioBERT) and scientific text processing (SciBERT) could be effectively applied to media analysis tasks.

This work has important implications for Filipino information retrieval, suggesting that models pre-trained on general multilingual corpora could benefit from additional adaptation to Filipino literary text. The principle that domain-specific pre-training enhances performance when the pre-training and

target domains share structural similarities indicates potential pathways for optimizing XLM-RoBERTa specifically for Filipino literature retrieval tasks.

2.6 Scoring Functions and Ranking Mechanisms

The effectiveness of hybrid retrieval systems depends critically on sophisticated scoring functions that appropriately weight multiple signals. Yoo et al. (2024) addressed this challenge in the context of abstractive summarization, proposing a hierarchical supervision method that jointly performs summary-level and sentence-level supervision. Their approach included intra- and inter-sentence ranking losses, enabling more detailed evaluation of content quality. While developed for summarization, their hierarchical scoring framework offered insights applicable to information retrieval, particularly regarding the importance of sentence-level detail in relevance assessment.

The concept of penalty factors in scoring functions was implicitly demonstrated by Bation et al. (2017), whose document classification system showed improved performance when stopword-heavy sentences received reduced weighting. This suggested that effective retrieval systems should incorporate penalty mechanisms for low-information content, a principle particularly relevant when processing documents containing substantial boilerplate or formulaic text.

2.7 Thematic Retrieval and Topic Modeling

While lexical matching and semantic similarity form the foundation of retrieval systems, thematic understanding represents a higher-level capability increasingly recognized as essential for literature-based applications. Ignaco and Ballera (2025) demonstrated this through their LDA-based Bible verse search system, which retrieved verses based on topical relevance rather than exact word matches. Their system achieved 33-67% accuracy in finding thematically relevant verses that

traditional keyword search completely missed (0% accuracy). Though retrieval time increased from 0.04 to 17-23 seconds, the study established that statistical topic modeling could uncover meaningful thematic connections invisible to keyword-based approaches. This work highlighted the fundamental trade-off between retrieval speed and thematic depth, a consideration essential for literature-focused retrieval systems.

The integration of transformer-based embeddings with topic modeling was explored by Aamir et al. (2025), who developed a framework combining BERTopic, XLM-R, and GPT for Urdu text analysis. Their approach achieved a coherence improvement of 0.05 and diversity score of 0.87 over traditional methods like LDA and NMF, demonstrating that transformer embeddings could capture contextual details and grammatical intricacies that statistical methods missed. Significantly, their work on Urdu—another low-resource language with complex morphology suggested pathways for applying similar techniques to Filipino text analysis.

Gaikwad et al. (2024) synthesized recent advances in document clustering, emphasizing that BERT embeddings, when applied as feature vectors, effectively captured intricate syntactical and semantic relationships for clustering tasks. Their analysis of BERTopic, which combines BERT embeddings with topic modeling, demonstrated comprehensive syntactical and semantic understanding of document content. The study established that transformer-based embeddings enabled more detailed thematic organization than traditional bag-of-words approaches, though the effectiveness depended on appropriate preprocessing and feature extraction strategies.

For Filipino literature specifically, thematic analysis has traditionally focused on canonical works such as *Noli Me Tangere* and *El Filibusterismo*. Rizal (1887, 1891) embedded recurring themes in these novels including religion, power, radicalism versus incrementalism, education, family

and honor, sacrifice, privilege, isolation, and revenge. These established thematic categories provide a framework for evaluating whether retrieval systems can capture conceptual relationships beyond surface-level lexical matching a capability essential for supporting literary analysis and education.

2.8 The Research Gap

The reviewed literature establishes a clear progression from basic keyword matching to sophisticated hybrid retrieval systems that integrate lexical, semantic, and thematic dimensions. However, several critical gaps remain, particularly regarding Filipino/Tagalog applications:

First, while XLM-RoBERTa has demonstrated strong performance on Filipino text classification and sentiment analysis tasks, its application to passage-level retrieval from literary texts remains unexplored. The code-switching capabilities documented by Cosme and De Leon (2024) and Salve and Tubil (2025) suggest potential for handling Filipino literature, but require validation in retrieval contexts.

Second, existing studies of hybrid retrieval systems (Gao et al., 2021; Kuzi et al., 2020) have focused primarily on high-resource languages and general document collections. The specific challenges of retrieving from Filipino literary works which exhibit distinctive linguistic features including formal/archaic language and culturally-specific references have not been systematically addressed.

Third, while thematic retrieval has been explored for religious texts (Ignacio & Ballera, 2025) and other domains, its application to Filipino literature analysis remains underdeveloped. The established thematic frameworks for works like *Noli Me Tangere* provide evaluation criteria, but no

existing system has demonstrated the ability to retrieve passages based on thematic similarity using Filipino-capable embeddings.

Fourth, the preprocessing requirements for Filipino text particularly regarding stopword handling, frequency-based validation, and deduplication have been examined in isolation (Bation et al., 2017; Kalykulova & Alzhanov, 2025) but not integrated into a comprehensive retrieval pipeline optimized for literary texts.

Fifth, while domain-adaptive pre-training has proven effective for specialized tasks in high-resource languages (Krieger et al., 2022), its application to low-resource language retrieval systems remains unexplored. The potential benefits of adapting multilingual models specifically to Filipino literary text characteristics have not been empirically validated.

Finally, scoring functions that appropriately balance lexical overlap, semantic similarity, and passage quality for Filipino text remain undeveloped. While general principles have been established (Yoo et al., 2024), their adaptation to Filipino linguistic characteristics requires empirical validation.

2.9 Theoretical Framework

The present study is grounded in three interconnected theoretical frameworks that collectively inform the design and implementation of the hybrid information retrieval system:

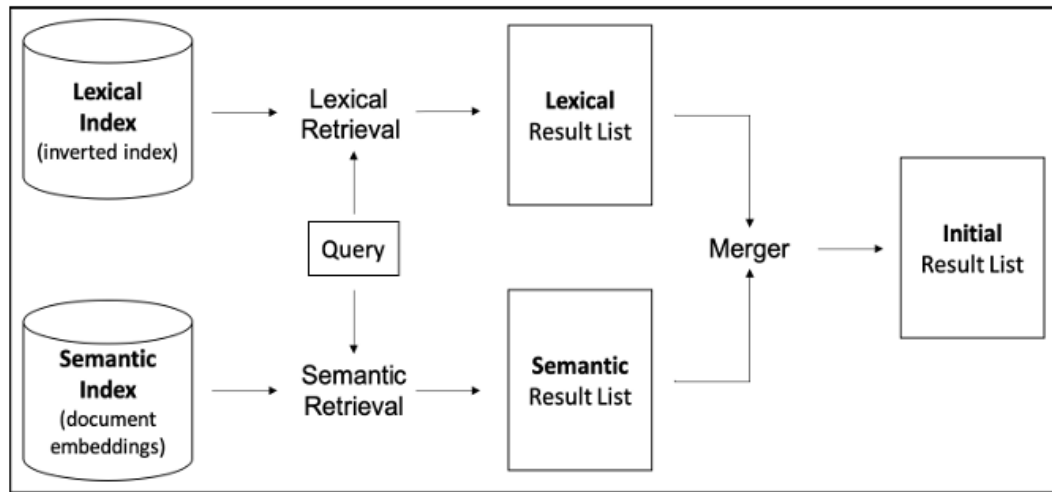


Figure 1: Theoretical Framework of a Hybrid Document Retrieval System Leveraging Semantic and Lexical Matching Based on Kuzi et al. (2020)

2.9.1 Hybrid Retrieval Theory

Based on the work of Gao et al. (2021) and Kuzi et al. (2020), Hybrid Retrieval Theory proposes that optimal information retrieval emerges from the integration of lexical and semantic approaches rather than their competition. Lexical methods excel at precise matching and handle out-of-vocabulary terms effectively, while semantic methods capture conceptual relationships and synonymy. The theory suggests that residual-based learning where semantic models are explicitly trained to capture information that lexical models miss produces superior retrieval performance compared to either approach alone. This framework guides the study's hybrid architecture, which

combines TF-IDF-based lexical matching with XLM-RoBERTa semantic embeddings through a weighted scoring function.

2.9.2. Cross-lingual Transfer Learning Theory

Drawing from the multilingual NLP research of Conneau et al. (2019), Imperial (2021), Cosme and De Leon (2024), and Salve and Tubil (2025), Cross-lingual Transfer Learning Theory explains how pre-trained multilingual models develop language-agnostic representations that transfer across typologically diverse languages. The theory proposes that transformer models trained on large multilingual corpora learn to map similar concepts to proximate regions in embedding space regardless of surface-level linguistic differences. For low-resource languages like Filipino/Tagalog, this enables using semantic understanding developed from high-resource languages. This framework justifies the selection of XLM-RoBERTa as the semantic encoder and informs the preprocessing strategies that maximize cross-lingual transfer effectiveness.

2.9.3 Thematic Coherence Theory

Synthesized from the work of Yoo et al. (2024), Ignaco and Ballera (2025), and Aamir et al. (2025), Thematic Coherence Theory proposes that document relevance operates at multiple levels lexical, semantic, and thematic each contributing distinct information to retrieval quality. Lexical relevance captures explicit term overlap; semantic relevance captures conceptual similarity through embeddings; thematic relevance captures higher-order topical relationships. The theory suggests that optimal retrieval requires explicit modeling of each level and appropriate weighting mechanisms that account for passage characteristics such as length, information density, and stopword presence. This framework guides the study's multi-component scoring function and thematic retrieval capabilities.

2.9.4 Integration of Theoretical Frameworks

These three theoretical frameworks interact in the present study. Hybrid Retrieval Theory provides the overarching architectural principle of integration. Cross-lingual Transfer Learning Theory enables the application of this architecture to Filipino text by using multilingual pre-training. Thematic Coherence Theory operationalizes the hybrid approach through specific scoring mechanisms that balance multiple relevance signals. Together, these frameworks establish a comprehensive theoretical foundation for developing and evaluating a Filipino literature retrieval system that advances beyond existing keyword-based approaches.

2.10 Synthesis of the Study

The reviewed literature reveals a clear trajectory of advancement in information retrieval systems, progressing from simple keyword matching to sophisticated hybrid approaches that integrate multiple forms of linguistic understanding. However, this progression has occurred primarily within high-resource language contexts, leaving significant gaps in the application of advanced retrieval techniques to Filipino/Tagalog literary texts.

2.10.1 Key Findings and Convergence

Several critical findings emerge from the literature synthesis:

Hybrid Superiority: The superiority of hybrid retrieval approaches over purely lexical or purely semantic methods has been conclusively established (Gao et al., 2021; Kuzi et al., 2020), providing strong justification for the architectural decisions in the present study.

XLM-RoBERTa Effectiveness: Multiple independent studies (Conneau et al., 2019; Imperial, 2021; Cosme & De Leon, 2024; Salve & Tubil, 2025; Timoneda & Vallejo Vera, 2023) have converged on XLM-RoBERTa as the optimal multilingual transformer architecture for Filipino text processing, supporting its selection for semantic encoding in this study.

Domain Adaptation Potential: Research on domain-adaptive pre-training (Krieger et al., 2022) has demonstrated that specialized adaptation of pre-trained models significantly enhances performance on domain-specific tasks, suggesting potential pathways for optimizing multilingual models for Filipino literary text retrieval.

Preprocessing Criticality: The importance of language-specific preprocessing—particularly stopword removal and term validation has been demonstrated across multiple contexts (Bation et al., 2017; Ladani & Desai, 2020; Kalykulova & Alzhanov, 2025), informing the preprocessing pipeline design.

Thematic Potential: Emerging research on topic modeling and thematic retrieval (Aamir et al., 2025; Gaikwad et al., 2024) suggests that embeddings can capture higher-order thematic relationships, though application to Filipino literature remains unexplored.

2.10.2 Research Gaps Addressed by This Study

The present study directly addresses several critical gaps identified in the literature:

Gap 1: Filipino Literary Retrieval While semantic retrieval has been extensively studied for general documents, news articles, and social media text, no existing research has examined passage-level retrieval from Filipino literary works. This study fills this gap by developing a system specifically optimized for canonical Filipino literature.

Gap 2: Hybrid Retrieval for Low-Resource Languages Existing hybrid retrieval systems have focused on high-resource languages with abundant training data and linguistic tools. This study extends hybrid retrieval methodology to Filipino, demonstrating how multilingual pre-trained models can enable sophisticated retrieval in low-resource contexts.

Gap 3: Thematic Retrieval in Filipino Literature While thematic frameworks exist for Filipino literature (Rizal, 1887, 1891), no computational system has attempted to retrieve passages based on thematic similarity using Filipino-capable embeddings. This study pioneers thematic retrieval for Filipino literary analysis.

Gap 4: Integrated Preprocessing for Filipino IR Previous studies have examined individual preprocessing techniques (stopword removal, frequency validation, deduplication) in isolation. This study integrates these techniques into a comprehensive pipeline specifically designed for Filipino literary text characteristics.

Gap 5: Empirical Validation of Scoring Functions While general principles for hybrid scoring exist, their application to Filipino text with its distinctive code-switching, formal register, and cultural references has not been empirically validated. This study provides systematic evaluation of scoring function components in the Filipino context.

Gap 6: Domain Adaptation for Filipino Retrieval While domain-adaptive pre-training has proven effective in specialized high-resource language tasks (Krieger et al., 2022), its potential application to low-resource language retrieval systems remains unexplored. This study lays groundwork for future research on adapting multilingual models specifically to Filipino literary domains.

2.10.3 Contribution to the Field

This study represents the logical next link in the chain of information retrieval research by:

Extending Proven Architectures: Applying the validated hybrid retrieval architecture of Gao et al. (2021) and Kuzi et al. (2020) to a new linguistic context, demonstrating its applicability beyond high-resource languages.

Leveraging Multilingual Capabilities: Building upon the Filipino NLP advances of Imperial (2021), Cosme and De Leon (2024), and Salve and Tubil (2025) by applying XLM-RoBERTa to a novel task domain (literary passage retrieval).

Advancing Thematic Understanding: Extending the topic modeling approaches of Ignaco and Ballera (2025) and Aamir et al. (2025) to Filipino literature, demonstrating that transformer embeddings can support thematic analysis in low-resource educational contexts.

Addressing Practical Needs: Responding to the documented challenges in Filipino literary education by providing a tool that supports close reading, thematic analysis, and passage discovery capabilities currently unavailable to educators and students.

2.10.4 Positioning Within the Research Landscape

The present study occupies a unique position at the intersection of several research trajectories: hybrid information retrieval, multilingual NLP, low-resource language processing, and digital humanities. It represents both an application of established principles to a new context and an extension of those principles to address context-specific challenges. By systematically evaluating lexical overlap, semantic similarity, and thematic relevance for Filipino literary texts, this study

provides empirical evidence that advances understanding of how retrieval systems should be designed for linguistically and culturally distinctive content.

The findings from this research have implications extending beyond Filipino literature to other low-resource literary traditions facing similar challenges in digital access and computational analysis. The methodological framework developed here combining language-specific preprocessing, multilingual embeddings, and multi-dimensional relevance scoring offers a template adaptable to other underserved literary contexts.

CHAPTER 3

METHODOLOGY

3.1 Research Design

This study employs a design and development research approach with comparative evaluation to create and assess a dual-formula hybrid information retrieval system for thematic analysis of José Rizal's *Noli Me Tangere* and *El Filibusterismo*. The research design consists of four primary phases: (1) domain-adaptive pre-training for semantic grounding validation, (2) system development with dual-formula architecture, (3) implementation and testing, and (4) comparative evaluation. This methodology directly addresses the research objectives by systematically developing, implementing, and evaluating a hybrid retrieval architecture that integrates XLM-RoBERTa semantic embeddings with weighted lexical matching and Filipino-specific preprocessing through two specialized scoring formulas, enhanced by domain-adaptive semantic validation.

The research is grounded in four theoretical frameworks established in Chapter 2: Hybrid Retrieval Theory (Gao et al., 2021; Kuzi et al., 2020), Cross-lingual Transfer Learning Theory (Conneau et al., 2019; Imperial, 2021), Thematic Coherence Theory (Yoo et al., 2024; Ignaco & Ballera, 2025; Aamir et al., 2025), and Domain-Adaptive Pre-training Theory (Krieger et al., 2022). These frameworks inform the architectural decisions, preprocessing strategies, dual-formula design, semantic validation mechanisms, and evaluation criteria throughout the methodology.

3.2 Preprocessing Pipeline

A systematic preprocessing pipeline transforms the raw summary texts into formats suitable for both lexical and semantic analysis. The pipeline implements language-specific preprocessing informed by the literature reviewed in chapter 2.

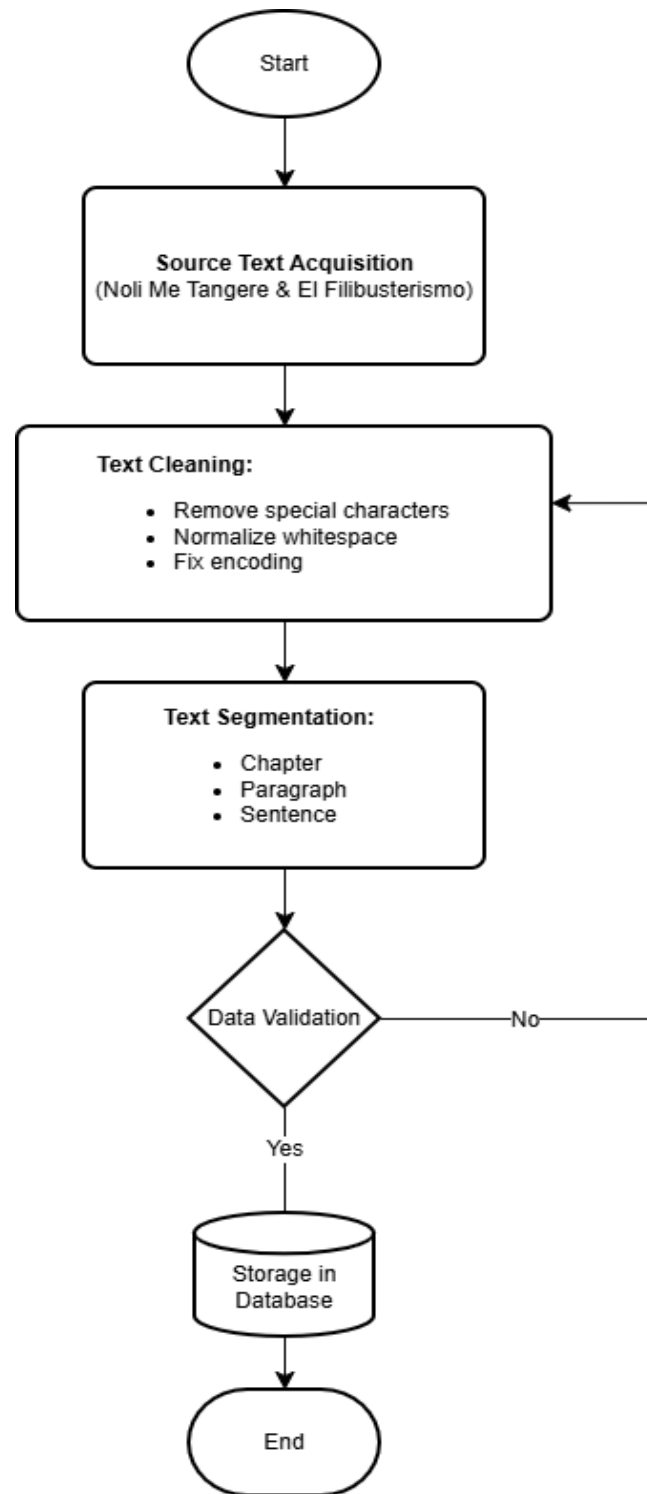


Figure 2: Data Collection and Preprocessing Pipeline

3.3 Conceptual Framework

The conceptual framework illustrates how the dual-formula hybrid information retrieval system integrates multiple theoretical foundations and technical components to enable thematic literary analysis of Rizal's novels. The framework operates through five interconnected layers with two distinct scoring formulas optimized for different retrieval contexts, enhanced by domain-adaptive semantic grounding validation.

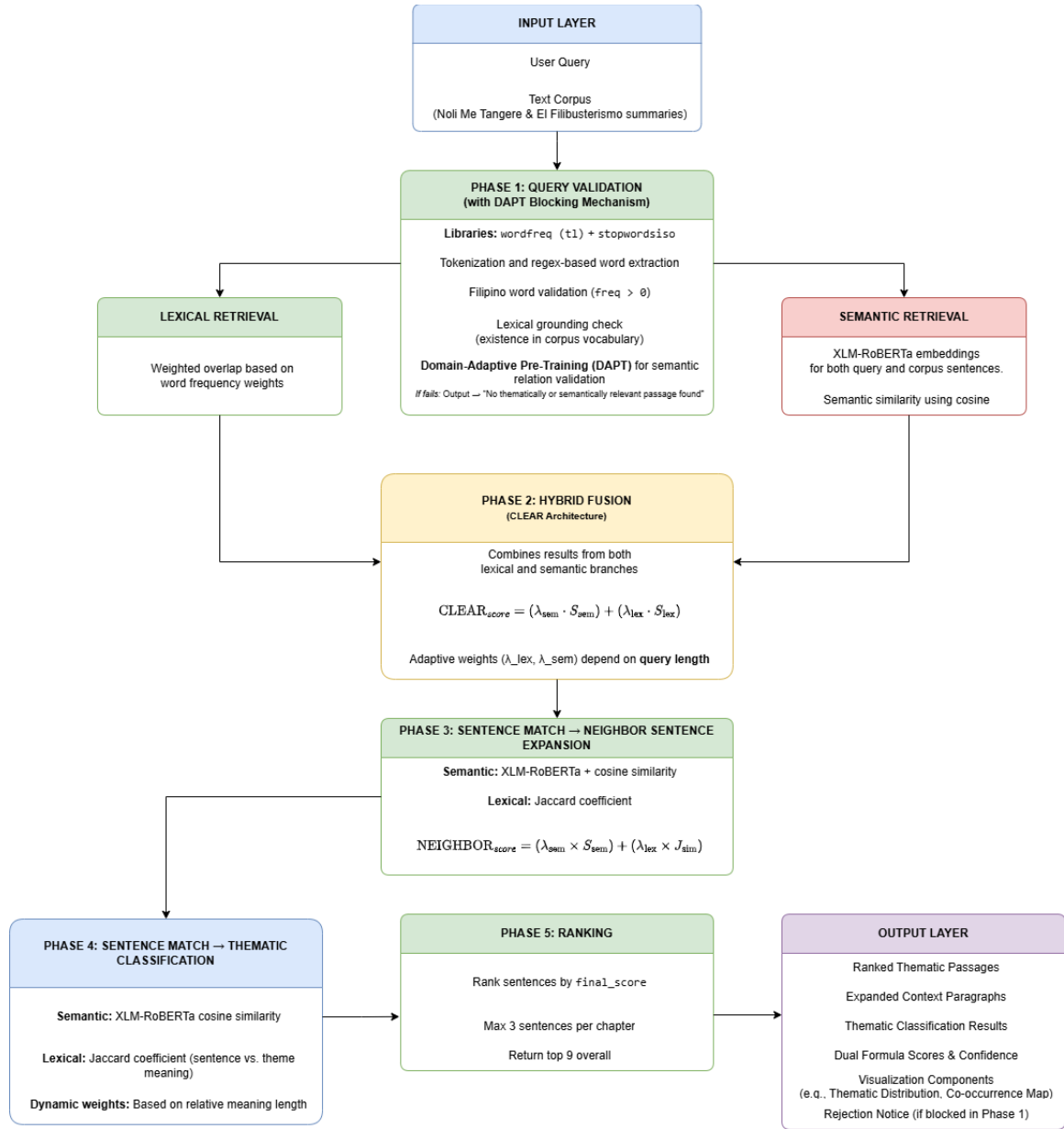


Figure 3: Conceptual Framework for the Hybrid Information Retrieval System Using XLM-RoBERTa for Thematic Literary Analysis of Rizal's Novels

3.3.1 Input Layer: Query and Corpus

The system receives two primary inputs:

User Query: Thematic queries in Filipino/Tagalog representing specific literary themes (e.g., "kolonyalismo," "edukasyon bilang pag-asa ni Ibarra," "katarungan," "kamatayan ni Basilio"). Queries undergo both linguistic validation to ensure they contain valid Filipino words with sufficient semantic content and domain-adaptive semantic grounding validation to verify meaningful connection to the corpus domain.

Text Corpus: Filipino language summary versions (*buod*) of *Noli Me Tangere* and *El Filibusterismo*, providing comprehensive coverage of major themes while maintaining linguistic consistency and manageable computational scope.

3.3.2 Processing Layer: Domain-Adaptive Semantic Validation

Drawing from Domain-Adaptive Pre-training Theory (Krieger et al., 2022), the system implements a novel semantic grounding validation layer that detects queries with no meaningful semantic relation to the corpus. This layer prevents the retrieval of weakly similar but thematically irrelevant passages, addressing a critical limitation of pure semantic similarity approaches.

Semantic Grounding Assessment:

The domain-adaptive validation computes the semantic alignment between the query and the entire corpus domain:

$$\text{DomainAlignment}(q, C) = \max_{p \in C} \text{CosineSim}(\mathbf{e}_q, \mathbf{e}_p)$$

where:

- $\mathbf{e}_q$ is the XLM-RoBERTa embedding of the query
- $\mathbf{e}_p$ are embeddings of all passages in corpus C
- Maximum similarity indicates strongest domain connection

Rejection Criteria:

Queries are rejected if:

1. **Semantic Mismatch:** $\text{DomainAlignment}(q, C) < \theta_{\text{domain}}$ where $\theta_{\text{domain}} = 0.30$
(empirically determined threshold)
2. **Lexical Absence:** All content words absent from corpus vocabulary
3. **Thematic Disconnect:** Query semantically distant from all predefined themes

Rejection Response:

Rather than returning weakly similar passages (which would constitute semantic hallucination), the system explicitly returns:

"No thematically or semantically relevant passage found. The query does not align with the literary content of the novels."

Example Application:

Query: "Kamatayan ni Basilio" (Death of Basilio)

Reality: Basilio does not die in the novel summaries.

System Behavior:

1. Computes DomainAlignment across all passages
2. Finds maximum semantic similarity = 0.24 (below 0.30 threshold)
3. Despite lexical matches for "Basilio," semantic embeddings indicate thematic mismatch
4. **Returns:** "No thematically or semantically relevant passage found."
5. **Does NOT hallucinate:** Results about "Basilio" or "sacrifice" that are semantically weak

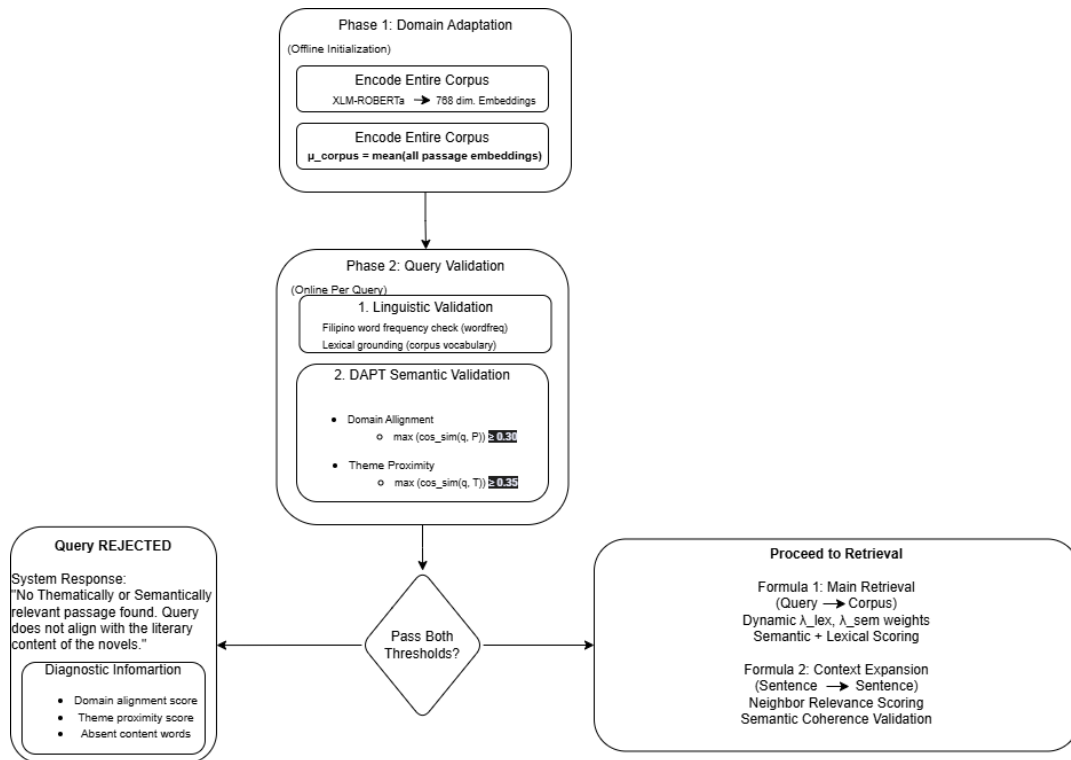


Figure 4: Domain-Adaptive Pre-training (DAPT) Validation Architecture for Semantic Grounding

This domain-adaptive validation implements the principle established by Krieger et al. (2022) that domain-specific adaptation improves task performance when the pre-training domain aligns with the target task. By explicitly modeling the semantic space of Rizal's novels, the system learns to distinguish between queries that belong to this domain versus those that do not, even when superficial lexical overlap exists.

3.3.3 Processing Layer: Dual-Formula Hybrid Retrieval Architecture

After passing semantic grounding validation, the processing layer implements a dual-formula hybrid retrieval architecture that employs two distinct scoring formulas optimized for different retrieval contexts:

Formula 1 (Main Retrieval): Query→Corpus matching with dynamic weights favoring lexical precision for short queries

Formula 2 (Neighbor Retrieval): Sentence→Sentence matching with dynamically adjusted weights that emphasize semantic continuity and rare-word importance for more accurate context expansion.

This dual-formula approach addresses the fundamental difference between:

- **Primary retrieval:** Finding passages directly relevant to user queries (requires precision and keyword alignment)
- **Context retrieval:** Finding thematically coherent neighboring sentences (requires semantic continuity)

The processing layer combines four parallel processing streams:

Preprocessing Pipeline: Applies text cleaning, Tagalog stopword removal (using stopwords-iso-tl), tokenization (using XLM-RoBERTa tokenizer), passage segmentation, and deduplication. This stream ensures text is properly formatted for both lexical and semantic analysis while removing noise and redundancy.

Semantic Component: Utilizes XLM-RoBERTa multilingual transformer model (sentence-transformers/paraphrase-xlm-r-multilingual-v1) to generate 768-dimensional contextual embeddings for both queries and passages. This component implements Cross-lingual Transfer Learning Theory, enabling semantic understanding of Filipino text through knowledge transfer from high-resource languages. Cosine similarity measures semantic relatedness between query and passage embeddings. The same model is used for both main and neighbor retrieval, but with different weighting schemes.

Lexical Component: Employs weighted term matching to quantify keyword overlap between queries and passages. Unlike traditional TF-IDF, the system implements stopwords-aware semantic weighting where each word receives a weight based on its corpus frequency and stopword status:

Frequency here means relative frequency, or how often a word appears compared to the total number of words in the corpus.

$$\text{frequency}(w_i) = \frac{\text{count}(w_i)}{\text{total tokens in corpus}}$$

- Stopwords: weight = 0.0
- Very common words (frequency > 0.001): weight = 0.3
- Medium frequency words (0.0001 < frequency ≤ 0.001): weight = 0.7

- Rare/content words (frequency ≤ 0.0001): weight = 1.0

This weighting preserves the precision advantages of exact-match retrieval while minimizing stopword influence.

Hybrid Integration: Combines semantic and lexical signals through two specialized multi-component scoring functions:

- **Main Formula:** Dynamically adjusts component weights based on query characteristics (length, stopword ratio), favoring lexical matching for short queries and semantic understanding for longer queries
- **Neighbor Formula:** Uses dynamically adjusted weights based on corpus word frequency and contextual relevance to assess sentence-to-sentence coherence. This adaptive weighting emphasizes semantically informative and rare words, enabling finer-grained thematic alignment between neighboring sentences.

This dual-formula integration operationalizes Hybrid Retrieval Theory by explicitly modeling complementary retrieval signals while adapting the balance to retrieval context.

3.3.4 Analysis Layer: Thematic Classification and Context Expansion

The analysis layer enriches retrieved passages with thematic information and applies neighbor-formula-based context expansion:

Thematic Classification: Matches retrieved passages against predefined themes from established literary sources (GradeSaver, LitCharts, Francisco) using semantic similarity. Passages

scoring above threshold confidence (≥ 0.45) are tagged with relevant themes, enabling thematic analysis beyond simple passage retrieval.

Context Expansion with Neighbor Formula: For top-ranked passages, the system identifies adjacent sentences using Formula 2 (Neighbor Retrieval) to assess semantic coherence between the main sentence and its neighbors. Only sentences exceeding the neighbor relevance threshold (combined score ≥ 0.40) are included as context. This ensures that context expansion maintains thematic continuity rather than including arbitrary adjacent sentences.

Multi-Component Scoring: Applies final ranking based on weighted combination of semantic similarity, lexical overlap, and passage length normalization using Formula 1 (Main Retrieval). This implements Thematic Coherence Theory by assessing relevance at lexical, semantic, and structural levels.

3.3.5 Output Layer: Retrieval Results and Visualization

The output layer presents results with multiple analytical dimensions, including transparent display of both formulas' contributions:

Ranked Passages: Top-k most relevant passages with associated metadata including:

- Chapter location (number and title)
- Match type (exact, partial lexical, semantic)
- Confidence scores (semantic, lexical, final) computed using Formula 1
- Thematic tags with confidence levels
- Context sentences with Formula 2 neighbor scores

Dual-Formula Metrics Display: The system explicitly shows:

- Formula 1 weights for main retrieval (λ_{lex} and λ_{sem})

$$(\lambda_{\text{lex}}, \lambda_{\text{sem}}) = \begin{cases} (0.75, 0.25) & \text{if sentence_length} \leq 5 \\ (0.65, 0.35) & \text{if } 5 < \text{sentence_length} \leq 10 \\ (0.55, 0.45) & \text{if } 10 < \text{sentence_length} \leq 15 \\ (0.45, 0.55) & \text{if } 15 < \text{sentence_length} \leq 20 \\ (0.35, 0.65) & \text{if sentence_length} > 20 \end{cases}$$

- Formula 2 dynamic weights for neighbor retrieval

$$(\lambda_{\text{lex}}, \lambda_{\text{sem}}) = \begin{cases} (0.30, 0.70) & \text{if length_ratio} \geq 1.5 \text{ (much longer)} \\ (0.40, 0.60) & \text{if } 1.2 \leq \text{length_ratio} < 1.5 \text{ (longer)} \\ (0.50, 0.50) & \text{if } 0.8 \leq \text{length_ratio} < 1.2 \text{ (similar)} \\ (0.60, 0.40) & \text{if } 0.5 \leq \text{length_ratio} < 0.8 \text{ (shorter)} \\ (0.70, 0.30) & \text{if length_ratio} < 0.5 \text{ (much shorter)} \end{cases}$$

- Individual semantic and lexical scores for each neighbor sentence
- Combined neighbor scores showing how Formula 2 evaluates context relevance

Performance Metrics: Quantitative measures including precision, recall, F1-score, and component contribution analysis (semantic vs. lexical vs. hybrid performance for both formulas).

Thematic Distribution: Visualization of how themes distribute across both novels, identifying thematic concentration patterns and chapter-level prevalence.

Co-occurrence Analysis: Identification and visualization of passages where multiple themes intersect, revealing thematic relationships and complexity in Rizal's social commentary.

Rejection Notifications: For queries failing domain-adaptive semantic validation, clear explanation of why no results were returned, preventing user confusion from weak or hallucinated matches.

3.3.6 Theoretical Integration

The conceptual framework integrates the four theoretical foundations as follows:

Domain-Adaptive Pre-training Theory provides the foundation for semantic grounding validation, ensuring that the system has been adapted to understand the specific domain of Rizal's novels. This prevents retrieval of passages that are superficially similar but thematically irrelevant, implementing Krieger et al.'s (2022) principle that domain adaptation improves performance when pre-training and target domains align.

Hybrid Retrieval Theory guides the architectural principle of integrating lexical and semantic retrieval through two specialized, context-aware formulas rather than relying on a single uniform weighting scheme. This dual-formula design acknowledges that the optimal balance between lexical precision and semantic understanding varies with retrieval context—direct query matching (Formula 1) demands adaptive term weighting for relevance, while contextual coherence assessment (Formula 2) benefits from dynamic, frequency-based weighting to preserve thematic flow across sentences.

Cross-lingual Transfer Learning Theory enables both semantic components to function effectively on Filipino text despite XLM-RoBERTa's limited Filipino-specific training data. The model's

multilingual pre-training creates shared semantic spaces where concepts cluster together regardless of language, supporting both main and neighbor retrieval.

Thematic Coherence Theory informs the hierarchical relevance assessment across lexical (keyword overlap), semantic (conceptual similarity), and thematic (topic classification) levels. The framework recognizes that literary relevance operates at multiple dimensions beyond surface-level matching, with Formula 1 optimizing for query-document relevance and Formula 2 optimizing for discourse-level coherence.

3.2.7 Framework Validation

The conceptual framework's effectiveness is validated through:

Domain-Adaptive Validation Testing: Evaluation of semantic grounding rejection for known-negative queries (queries about events/characters not in the novels) to measure false positive prevention.

Comparative Evaluation: Hybrid system performance compared against keyword-only baseline to demonstrate value of semantic integration.

Ablation Studies: Individual component testing (semantic-only, lexical-only, main formula vs. neighbor formula, hybrid without penalties, with/without domain validation) to validate architectural decisions.

Formula Effectiveness Analysis: Evaluation of whether Formula 2 produces more coherent context than using Formula 1 for neighbor retrieval.

Expert Review: Qualitative assessment by literature educators to confirm thematic relevance and practical utility.

Distribution Analysis: Empirical examination of thematic patterns to demonstrate capabilities beyond traditional close-reading.

This conceptual framework provides the organizing structure for the detailed methodology that follows, showing how theoretical foundations translate into practical dual-formula system design with domain-adaptive semantic validation and evaluation procedures.

3.4. Textual Corpus

The study utilizes Filipino language summary versions (*buod*) of both novels rather than the complete original texts. This decision was made based on several practical and linguistic considerations:

Noli Me Tangere Summary: Sourced from PinoyCollection.com (2017), providing comprehensive coverage of the novel's 64 chapters with major plot points and themes preserved in sentence-level segmentation.

El Filibusterismo Summary: Sourced from Noypi.com.ph (2019), covering all 39 chapters with thematic elements preserved in sentence-level segmentation.

3.4.1 Rationale for Using Summary Versions

The use of summaries instead of full-length texts addresses several challenges:

Linguistic Consistency: The original novels contain mixed Spanish-Tagalog content reflecting the linguistic reality of late 19th-century Philippines. These summaries, written in contemporary Filipino, provide linguistic consistency suitable for Tagalog-focused NLP processing using XLM-RoBERTa without requiring complex multilingual historical language handling.

Manageable Computational Scope: Summary versions offer manageable text lengths (approximately 15,000-20,000 words combined) suitable for iterative development and testing within thesis timeframe constraints, while full texts would exceed 200,000 words.

Thematic Representation: Educational summaries retain core thematic content, narrative structure, and key passages that exemplify major themes, making them appropriate for thematic analysis while reducing computational overhead.

Educational Relevance: These summaries represent materials actually used by Filipino high school students studying Rizal's works, increasing the practical relevance of the retrieval system for its intended educational audience.

Domain-Adaptive Pre-training Basis: The focused corpus scope enables effective domain adaptation, where the system learns the semantic space of Rizal's novels specifically, facilitating semantic grounding validation for query rejection.

3.4.2 Thematic Data Sources

Themes for evaluation were derived from established literary analysis sources rather than automated discovery:

Noli Me Tangere themes: Sourced from GradeSaver (2024) and Francisco (2025), including themes such as colonialism (*kolonyalismo*), education (*edukasyon*), religion (*relihiyon*), social inequality (*hindi pagkakapantay-pantay sa lipunan*), and corruption (*katiwalian*).

El Filibusterismo themes: Sourced from LitCharts (2025) and GradeSaver (2023), including themes such as justice (*katarungan*), revolution (*rebolusyon*), violence versus reform (*karahasan laban sa reporma*), greed (*kasakiman*), and disillusionment (*pagkabigo*).

These predefined themes serve as queries for the hybrid retrieval system and provide the basis for evaluation, acknowledging the study's delimitation that it does not perform automated theme discovery. The themes also serve as domain markers for semantic grounding validation, establishing the expected semantic space of valid queries.

3.4.3 Domain-Adaptive Corpus Representation

Following Krieger et al. (2022), the corpus is represented in a domain-adapted semantic space through:

Corpus Embedding Generation: All passages from both novels are encoded using XLM-RoBERTa to create a 768-dimensional semantic representation of the domain.

Domain Centroid Computation: The mean embedding vector across all passages establishes the semantic center of the Rizal novels domain:

$$\mathbf{c}_{\text{domain}} = \frac{1}{|C|} \sum_{p \in C} \mathbf{e}_p$$

where C is the set of all corpus passages and $\mathbf{e}_p$ are passage embeddings.

Semantic Boundary Calibration: The minimum and maximum semantic similarities between the domain centroid and actual passages define the expected range of valid queries. Queries falling significantly below this range (by more than one standard deviation) are flagged for potential rejection.

This domain representation enables the system to distinguish between queries that belong to the novels' thematic space versus those that reference events, characters, or concepts not present in the corpus.

3.4.4 Dataset Preparation and Annotation Process

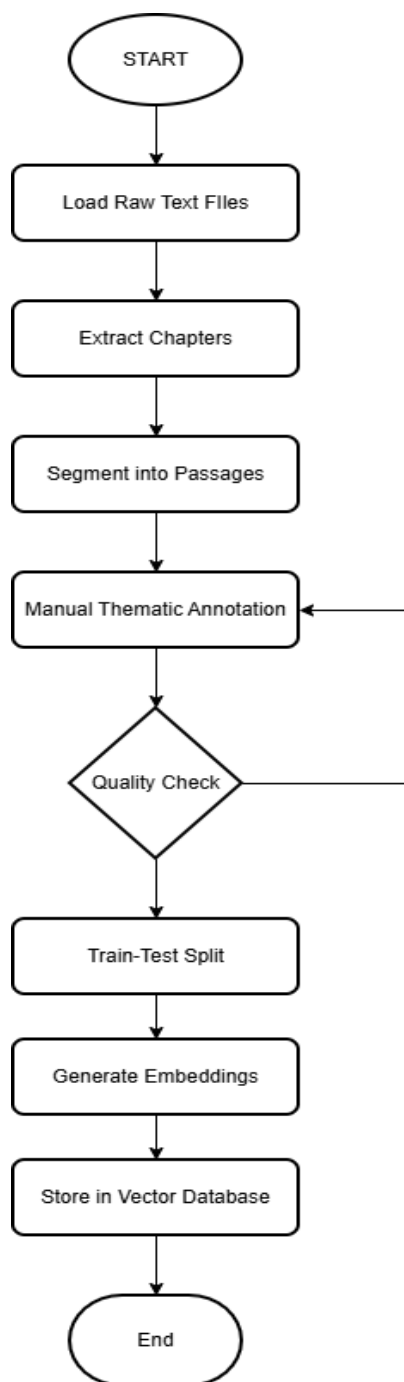


Figure 5: Dataset Preparation and Annotation Process

3.5 Text Cleaning and Normalization

Initial preprocessing steps standardize the text:

HTML and Special Character Removal: Stripping any HTML tags, URLs, or non-textual elements from the web-scraped summaries.

Case Normalization: Converting all text to lowercase to standardize word forms and reduce vocabulary size, following standard information retrieval practices (AlShammari, 2023).

Punctuation Handling: Removing punctuation marks except sentence-terminating periods, which are retained for sentence segmentation purposes.

Whitespace Normalization: Collapsing multiple spaces into single spaces and removing leading/trailing whitespace.

3.5.1 Tagalog Stopword Management

Common Tagalog words with minimal semantic value are identified using the stopwords-iso-tl package, which provides comprehensive coverage of approximately 200 Filipino/Tagalog stopwords including "ang," "mga," "si," "ng," "sa," "ay," "bilang," "ni," and "dahil."

The system implements a dual approach to stopwords handling:

- **For lexical scoring:** Stopwords are assigned semantic weight of 0.0, effectively removing their influence on lexical matching
- **Content words:** Receive weights of 0.3-1.0 based on their frequency characteristics

This approach effectively minimizes stopword influence on lexical matching while maintaining semantic understanding through neural embeddings. Stopword removal has been demonstrated to significantly improve Tagalog text processing performance, with Bation et al. (2017) achieving 92% accuracy in automatic document categorization when these stopwords were removed prior to feature extraction. Additionally, stopword removal can decrease corpus size by 35-45% while improving efficiency and accuracy (Ladani & Desai, 2020). The system balances these benefits by removing stopwords for lexical matching while preserving them in neural embeddings to maintain grammatical and semantic context.

Stopword Weighting Examples

Example 1: Query - "ang edukasyon ni Ibarra"

Word-by-Word Weight Assignment:

Table 3: Word-by-Word Weight Assignment (Example 1)

Word	Type	Weight	Contribution to Total
ang	Stopword	0	0
edukasyon	Content	0.7	0.7
ni	Stopword	0	0
Ibarra	Content	1	1
TOTAL QUERY WEIGHT	—	—	1.7

Stopword Effect:

- Original query: 4 words
- Stopwords: "ang", "ni" (50% of query)
- Effective content: "edukasyon" + "Ibarra"

- Stopwords contribute 0.0 to lexical scoring
- Only content words (weight 1.7) determine lexical overlap

Example 2: Query - "Edukasyon bilang pag-asa ni Ibarra"

Word-by-Word Weight Assignment:

Table 4: Word-by-Word Weight Assignment (Example 2)

Word	Type	Weight	Contribution to Total
Edukasyon	Content	0.7	0.7
bilang	Stopword	0.0	0.0
pag-asa	Content	0.7	0.7
ni	Stopword	0.0	0.0
Ibarra	Content	1.0	1.0
TOTAL QUERY WEIGHT	—	—	2.4

Stopword Effect:

- Original query: 5 words
- Stopwords: "bilang", "ni" (40% of query)
- Effective content: "Edukasyon" + "pag-asa" + "Ibarra"
- Stopwords contribute 0.0 to lexical scoring
- Only content words (weight 2.4) determine lexical overlap

These examples demonstrate how stopwords like "ang," "ni," and "bilang" receive zero weight in lexical matching, ensuring that only semantically meaningful terms contribute to relevance scoring, while the complete query with stopwords is preserved for XLM-RoBERTa semantic embeddings to maintain grammatical context.

3.5.2 Tokenization for XLM-RoBERTa

Text is tokenized using the pre-trained tokenizer associated with the xlm-roberta-base model via the SentenceTransformer wrapper. This tokenizer employs SentencePiece subword tokenization, which handles out-of-vocabulary words by breaking them into meaningful subword units. This approach is essential for processing Filipino text, which may contain:

- Informal spellings
- Morphological variations not fully represented in the model's training vocabulary
- Hyphenated compounds (e.g., "pag-asa")

The tokenizer configuration matches that used during XLM-RoBERTa pre-training to ensure compatibility with the embedding model. The system uses a custom word pattern regex `[0-9a-zA-ZÀ-ÿñÑ]+(?:-[0-9a-zA-ZÀ-ÿñÑ]+)*` that preserves hyphenated tokens as single units, which is crucial for Filipino compound words.

3.5.3 Passage Segmentation Strategy

To enable passage-level retrieval as specified in the scope, the corpus is segmented into discrete passages. A passage is defined as a single sentence from the chapter summaries. This sentence-level granularity was chosen because:

- Summary sentences are typically substantive and thematically coherent
- Sentence boundaries provide natural semantic units
- Fine-grained retrieval enables precise theme localization
- Context expansion (Section 3.4.6) can aggregate multiple sentences when needed for broader understanding

Each passage is indexed with metadata including:

- `chapter_number`: Integer chapter identifier
- `sentence_number`: Integer sentence identifier within chapter
- `chapter_title`: String chapter title
- `sentence_text`: String passage content
- `sentence_word_count`: Integer word count for length penalty computation

3.5.4 Deduplication

Following preprocessing and segmentation, near-duplicate passages are identified and removed to prevent redundant information from skewing retrieval results. The system implements global deduplication that tracks used passages across all retrieval operations:

Initial Corpus Deduplication: Near-duplicate sentences identified using TF-IDF cosine similarity with threshold of 0.90. Only the first occurrence is retained.

Runtime Deduplication: Once a passage is retrieved for a query, it and its context sentences are marked as "used" (stored in `self.used_passages[book_key]` set) and excluded from subsequent results for that query. This ensures result diversity and prevents the same content from appearing multiple times in different forms.

Passage ID System: Each passage receives a unique identifier tuple (`chapter_number`, `sentence_number`) for efficient tracking.

This strict deduplication addresses redundancy challenges identified in the literature (Hendrickx et al., 2009) and ensures clean, non-repetitive result sets.

3.5.5 Context Expansion with Neighbor Formula

For retrieved passages, the system identifies adjacent sentences that provide thematic context using Formula 2 (Neighbor Retrieval). Context expansion operates as follows:

Boundary Detection: Identify sentences immediately before and after the retrieved passage within the same chapter.

Neighbor Scoring: For each adjacent sentence, compute:

- Semantic similarity between main sentence and neighbor using XLM-RoBERTa embeddings
- Lexical similarity using Jaccard coefficient (intersection over union of words)
- Combined neighbor score using Formula 2:

$$\text{neighbor_score} = (\lambda_{\text{sem}} \times \text{neighbor_semantic}) + (\lambda_{\text{lex}} \times \text{neighbor_lexical})$$

Relevance Filtering: Only sentences exceeding relevance threshold (0.40 combined neighbor score) are included as context. This higher threshold (compared to 0.30 for main retrieval) ensures context sentences are genuinely relevant.

Distance Limitation: Expand in both directions without sentence limits, stopping early if:

- Irrelevant sentence encountered (breaks thematic continuity)
- Chapter boundary reached
- Previously used passage encountered (deduplication)

Metadata Preservation: Track which context sentences are thematically relevant vs. included for narrative continuity, along with their neighbor scores.

This context expansion enables users to understand how themes manifest in surrounding narrative, supporting close reading and interpretive analysis. The use of Formula 2 ensures context expansion maintains semantic coherence rather than including arbitrary adjacent text.

3.6 System Architecture

The dual-formula hybrid information retrieval system integrates lexical and semantic retrieval components through a multi-stage architecture with two specialized scoring formulas and domain-adaptive semantic validation.

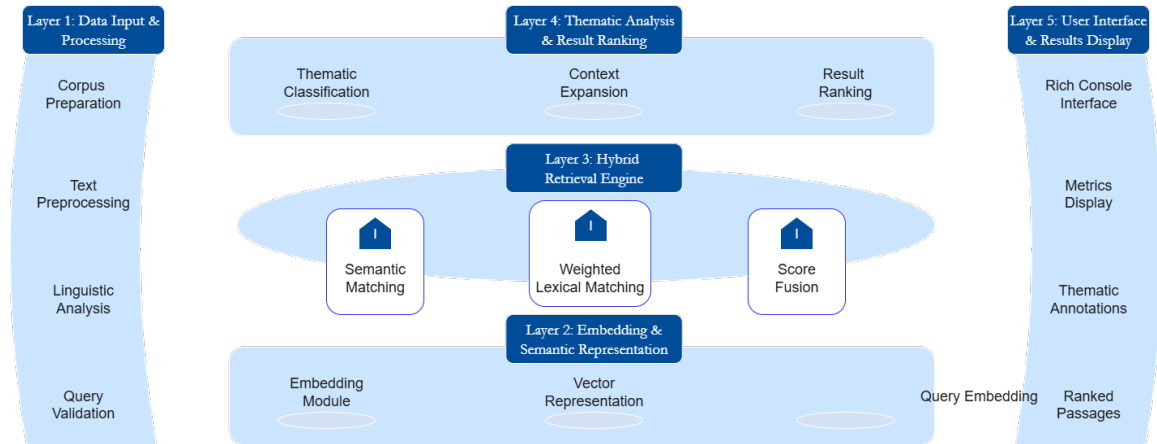


Figure 6: System Architecture of the Hybrid Information Retrieval System Using XLM-RoBERTa for Rizal's Novels

3.6.1 Overall Architecture

The system comprises six interconnected modules:

Data Ingestion Module: Loads preprocessed passages from CSV files (`noli_chapters.csv`, `elfili_chapters.csv`, `noli_themes.csv`, `elfili_themes.csv`) with metadata including chapter number, sentence number, chapter title, and sentence text.

Embedding Generation Module: Uses SentenceTransformer wrapper for XLM-RoBERTa (`sentence-transformers/paraphrase-xlm-r-multilingual-v1`) to generate contextual embeddings for all passages and themes during initialization. Embeddings are cached in memory for efficient retrieval. Combined text (chapter title + sentence text) is embedded for main retrieval. Domain centroid and semantic boundaries are computed for validation purposes.

Query Processing Module:

- Validates queries for Filipino word content using wordfreq library
- Analyzes word frequencies and stopword ratios
- Assigns semantic weights to query words
- Generates query embeddings using XLM-RoBERTa

Domain-Adaptive Semantic Validation Module:

- Computes semantic alignment between query and corpus domain
- Evaluates lexical presence of content words in corpus vocabulary
- Assesses thematic distance from predefined themes
- Applies rejection criteria for queries with insufficient domain grounding
- Returns explicit rejection message when query fails validation

Dual-Formula Hybrid Retrieval Module:

- Executes parallel semantic and lexical retrieval using Formula 1 for main passage retrieval
- Computes multi-component scores with dynamic weights
- Applies deduplication filters
- Expands context using Formula 2 for neighbor relevance assessment

Ranking and Presentation Module:

- Orders passages by Formula 1 hybrid relevance scores
- Expands context for top results using Formula 2
- Performs thematic classification

- Formats results with visualization showing both formulas' contributions
- Displays rejection notifications for failed queries

Hybrid Retrieval System Architecture

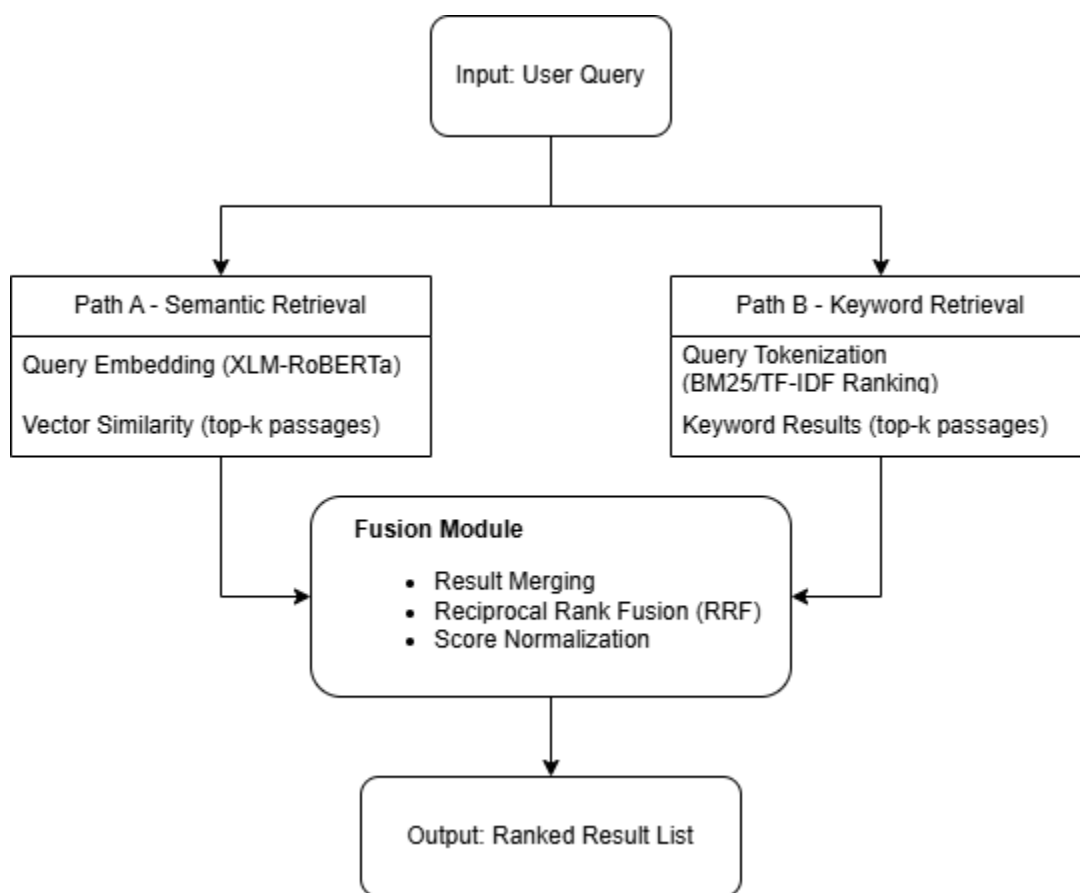


Figure 7: Hybrid Retrieval System Architecture (Detailed Component View)

3.6.2 Query Validation Component

Before retrieval begins, queries undergo linguistic and domain-adaptive semantic validation to ensure they contain valid Filipino words and meaningful connection to the corpus domain.

Phase 1: Linguistic Validation

Word Frequency Validation: Uses the wordfreq library to check if query words exist in Filipino language with minimum frequency threshold of 1×10^{-8} . This threshold filters out random character sequences and non-Filipino words while accepting rare but valid terms.

Stopword Analysis: Computes the ratio of stopwords to total words. Since the stopwords ratio is fixed at 1.0 (all words are stopwords), this metric becomes non-informative and is excluded from affecting Formula 1 retrieval weights.

Valid Word Ratio: Requires 100% of query words to be valid Filipino words. Queries failing this threshold are rejected with diagnostic feedback showing which words were not recognized.

Lexical Grounding Check: Validates that content words (non-stopwords) exist in the corpus vocabulary built from both novels and theme files. Queries with zero lexical grounding proceed to Phase 2 for semantic validation rather than immediate rejection.

Phase 2: Domain-Adaptive Semantic Validation

Drawing from Krieger et al. (2022), this validation layer implements domain-specific semantic grounding assessment:

Domain Alignment Computation: Calculate maximum semantic similarity between query embedding and all corpus passage embeddings:

$$\text{DomainAlignment}(q, C) = \max_{p \in C} \text{CosineSim}(\mathbf{e}_q, \mathbf{e}_p)$$

Semantic Threshold Application: If $\text{DomainAlignment}(q, C) < 0.30$, the query is considered semantically disconnected from the corpus domain.

Thematic Distance Assessment: Compute semantic similarity to all predefined themes:

$$\text{ThemeProximity}(q, T) = \max_{t \in T} \text{CosineSim}(\mathbf{e}_q, \mathbf{e}_t)$$

where T is the set of theme embeddings.

Combined Rejection Criteria: A query is rejected if:

$$\text{DomainAlignment}(q, C) < 0.30 \wedge \text{ThemeProximity}(q, T) < 0.35$$

This ensures that queries must be either semantically aligned with actual passages OR thematically related to known themes to proceed to retrieval.

Rejection Response: When a query fails validation, the system returns:

❑ "No thematically or semantically relevant passage found. The query does not align with the Krieger literary content of the novels."

❑ Along with diagnostic information:

- Maximum domain alignment score achieved
- Maximum theme proximity score achieved
- Content words that were lexically absent
- Suggestion to reformulate query or verify its relevance to the novels

Example Validation Flow:

Query: "Kamatayan ni Basilio" (Death of Basilio)

1. Linguistic Validation: ✓ PASS

- All words valid Filipino
- Content words: {kamatayan, Basilio}
- Lexical grounding: "Basilio" present in corpus

2. Semantic Validation:

- Domain alignment: 0.24 (< 0.30 threshold)
- Theme proximity: 0.28 (< 0.35 threshold)
- **Result:** ✗ REJECT

3. System Response:

❑ No thematically or semantically relevant passage found.

Domain alignment: 24.0% (below 30.0% threshold)

Theme proximity: 28.0% (below 35.0% threshold)

The query may reference events or concepts not present in the novel summaries.

❑ This domain-adaptive validation prevents the system from hallucinating results about Basilio's death (which does not occur in the summaries) by recognizing the semantic mismatch despite lexical presence of "Basilio."

3.6.3 Multi-Sentence Query Validation Protocol

For queries containing multiple sentences (detected by period-separated segments), the system applies sequential validation:

Validation Sequence:

1. Parse input into sentence units using period (.) delimiter
2. Validate first sentence through complete validation pipeline:
 - Linguistic validation (Phase 1)
 - Domain-adaptive semantic validation (Phase 2)
3. **Fail-Fast Mechanism:** If first sentence fails validation:
 - Halt processing immediately
 - Return error message:

❗"Sentence 1 is invalid due to [specific reason: semantic relation mismatch / domain misalignment / lexical absence]"

- ❗Do NOT process subsequent sentences
4. If first sentence passes, proceed to validate second sentence using same criteria
 5. Continue until all sentences validated or first failure encountered

Rationale: This sequential fail-fast approach prevents merging valid and invalid concepts in multi-sentence queries, maintaining result integrity. It ensures that semantically contradictory or contextually inappropriate queries are rejected before retrieval, even when embedded within longer multi-sentence input.

Example:

Input: "Kabaitan ni Padre Damaso. Edukasyon ni Ibarra."

Processing:

- Sentence 1: "Kabaitan ni Padre Damaso"
 - Semantic validation: Domain alignment = 0.19 (< 0.30)
 - Result: X REJECT
 - Output: "Sentence 1 is invalid due to semantic relation mismatch. Padre Damaso is portrayed as antagonistic in the novel, contradicting the attribution of virtue (kabaitan)."
- Sentence 2 is NOT processed

This protocol implements rigorous semantic grounding consistent with the domain-adaptive validation framework (Section 3.2.2).

3.6.4 Semantic Retrieval Component

The semantic component leverages XLM-RoBERTa to capture thematic similarity beyond exact keyword matching, implementing Cross-lingual Transfer Learning Theory. This component is used in both Formula 1 and Formula 2, but with different weighting schemes.

Model Selection:

The system uses sentence-transformers/paraphrase-xlm-r-multilingual-v1, a fine-tuned variant of xlm-roberta-base (110M parameters, trained on 100 languages) optimized for semantic similarity tasks. This model was chosen because:

- Pre-trained on 100+ languages including Filipino
- Fine-tuned specifically for semantic similarity using sentence-level contrastive learning

- Computationally efficient for CPU-based inference
- No fine-tuning required, following validated approaches for Filipino low-resource contexts (Imperial, 2021)

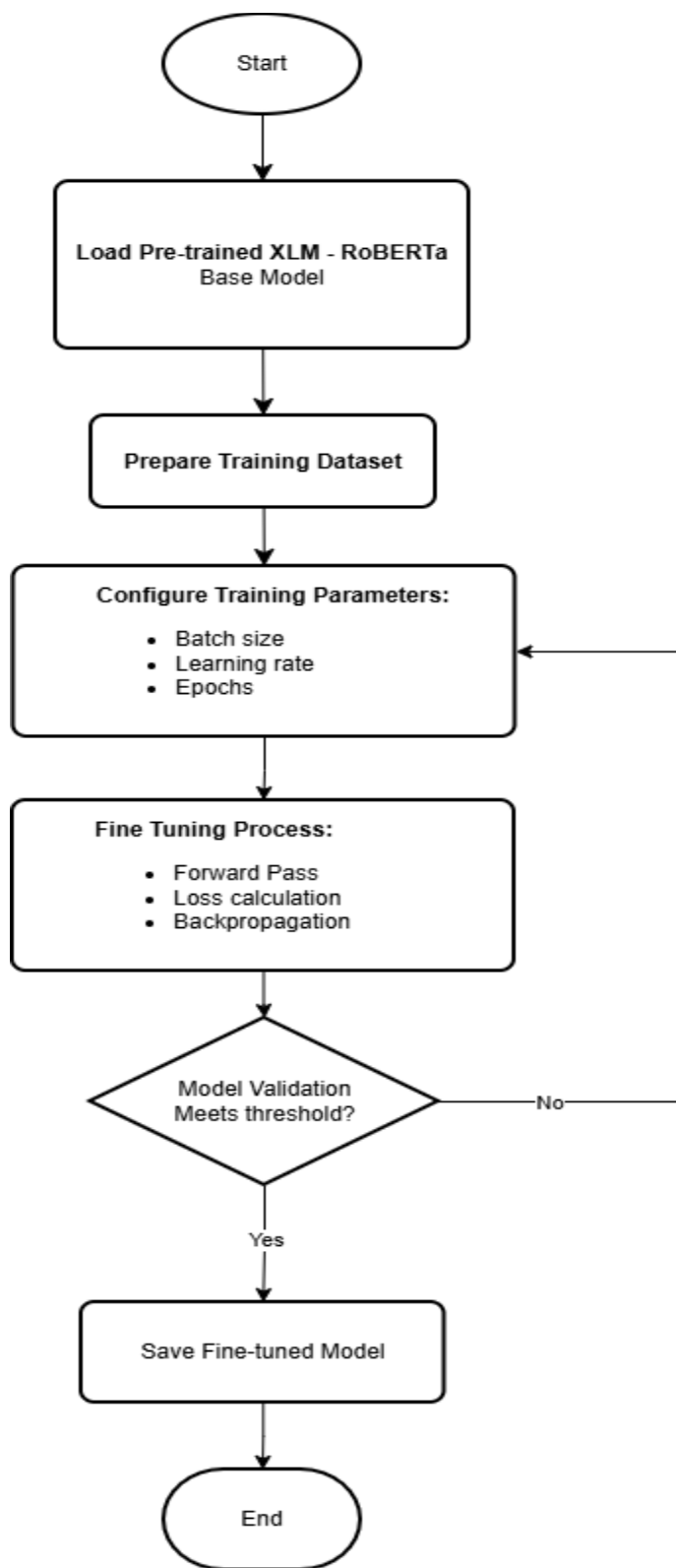


Figure 8: Model Training and Fine-Tuning Process

Embedding Generation: Each passage and query is encoded into a 768-dimensional dense vector representation. The encoding process:

$$\mathbf{e}_q = \text{XLM-RoBERTa}(\text{query})$$

$$\mathbf{e}_p = \text{XLM-RoBERTa}(\text{passage})$$

where $\mathbf{e}_q$ and $\mathbf{e}_p$ are embedding vectors in $\mathbb{R}^{768}$.

For main retrieval, passages are encoded with combined text (chapter title + sentence text) to provide additional context. For neighbor retrieval, only sentence text is encoded.

Sentence-Level Embedding Generation

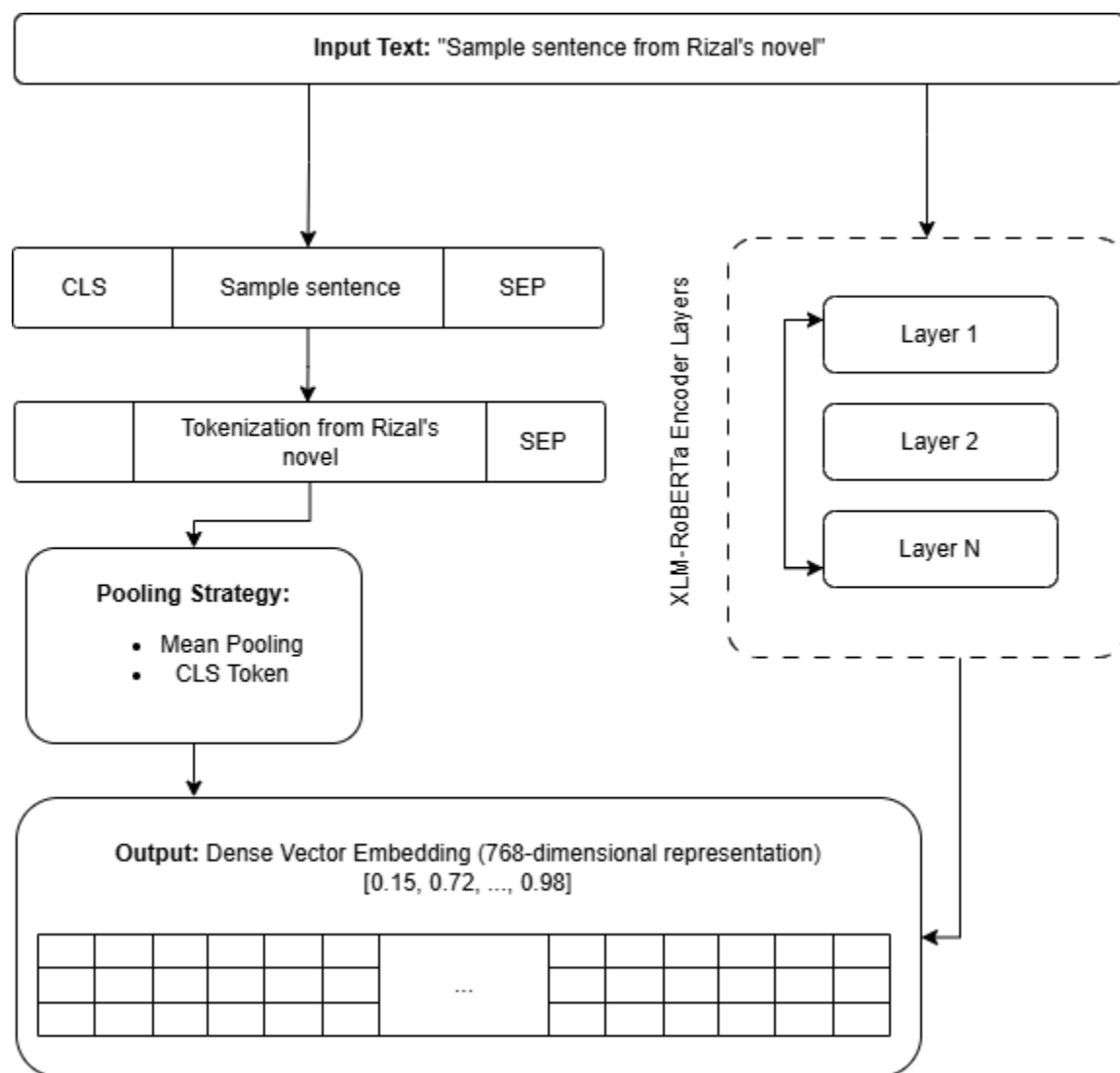


Figure 9: Sentence-Level Embedding Generation

Similarity Measurement: Cosine similarity quantifies semantic relatedness:

$$\text{CosineSim}(q, p) = \frac{\mathbf{e}_q \cdot \mathbf{e}_p}{|\mathbf{e}_q||\mathbf{e}_p|}$$

Cosine similarity ranges from -1 to 1, with values closer to 1 indicating greater semantic alignment. A minimum semantic threshold of 0.20 filters out clearly irrelevant passages before further processing in Formula 1.

According to Jain et al. (2017), Cosine similarity is fundamental in Vector Space Models (VSMs), where documents are represented as term-weighted vectors in an n-dimensional space. Cosine effectively captures directional similarity between document-query vectors, quantifying their proximity in semantic space regardless of vector magnitude, thereby enhancing retrieval precision in hybrid IR systems.

3.6.5 Lexical Retrieval Component

The lexical component captures explicit keyword overlap using stopwords-aware weighted term matching, preserving the precision advantages of exact-match retrieval while accommodating stopwords influence. This component is used in both formulas but with different implementations.jaccard

Semantic Weight Assignment: Each word in the query receives a semantic weight based on its linguistic properties determined by the wordfreq library and stopwords list:

$$w_{\text{word}} = \begin{cases} 0.0 & \text{if word is a stopwords} \\ 0.3 & \text{if frequency} > 0.001 \text{ (very common)} \\ 0.7 & \text{if frequency} > 0.0001 \text{ (medium)} \\ 1.0 & \text{otherwise (rare/content word)} \end{cases}$$

This weighting scheme ensures that content words (*edukasyon, katiwalian, katarungan*) contribute more to lexical scores than grammatical stopwords (*ang, mga, sa, bilang, ni*).

Weighted Lexical Overlap Score (Formula 1): For each query-passage pair in main retrieval:

$$\text{LexicalOverlap}(q, p) = \frac{\sum_{t \in q \cap p} w_t}{\sum_{t \in q} w_t}$$

where:

- w_t is the semantic weight of term t
- $q \cap p$ represents terms present in both query and passage
- The denominator normalizes by total query weight

Jaccard Lexical Similarity (Formula 2): In the neighbor-to-neighbor comparison for context expansion, similarity is computed using the Jaccard Coefficient rather than weighted overlap. According to Jain et al. (2017), the Jaccard coefficient measures set-based overlap without considering term weights, making it particularly suitable for assessing lexical coherence between adjacent sentences where the focus is on shared vocabulary rather than weighted importance:

$$\text{LexicalSim}_{\text{neighbor}}(m, n) = \frac{|W_m \cap W_n|}{|W_m \cup W_n|}$$

where:

- W_m is the set of words in the main sentence
- W_n is the set of words in the neighbor sentence
- Jaccard coefficient measures set overlap without weighted terms

The choice of Jaccard over weighted overlap for neighbor comparison is theoretically grounded: context expansion prioritizes lexical continuity and vocabulary cohesion rather than term importance hierarchies. Both Cosine and Jaccard similarity measures are fundamental in Vector Space Models (VSMs), where documents are represented as term-weighted vectors in an n -dimensional space (Jain et al., 2017). These measures effectively quantify document-query proximity, with Cosine capturing directional similarity in weighted semantic space and Jaccard emphasizing set-based overlap in unweighted lexical space—together enhancing retrieval precision and recall in hybrid IR systems.

3.6.6 Dual-Formula Hybrid Integration

The two retrieval components operate in parallel, with their outputs combined through two specialized formulas optimized for different retrieval contexts, directly addressing the research objectives.

Formula 1: Main Retrieval (Query→Corpus)

Purpose: Retrieve passages directly relevant to user queries with dynamic balancing of lexical precision and semantic understanding.

Dynamic Weight Computation: Weights are computed as a function of sentence (query) length (sentence_length):

$$(\lambda_{\text{lex}}, \lambda_{\text{sem}}) = \begin{cases} (0.75, 0.25) & \text{if sentence_length} \leq 5 \\ (0.65, 0.35) & \text{if } 5 < \text{sentence_length} \leq 10 \\ (0.55, 0.45) & \text{if } 10 < \text{sentence_length} \leq 15 \\ (0.45, 0.55) & \text{if } 15 < \text{sentence_length} \leq 20 \\ (0.35, 0.65) & \text{if sentence_length} > 20 \end{cases}$$

Final Score Computation (Formula 1):

$$\text{clear_score_base} = (\lambda_{\text{sem}} \times \text{semantic_score}) + (\lambda_{\text{lex}} \times \text{lexical_score})$$

Score Normalization:

$$\text{final_score} = \max(0.0, \min(1.0, \text{clear_score}))$$

Rationale: Longer queries provide more semantic context, justifying higher semantic weight.

Short queries (1-2 words) rely more on lexical precision. Since stopwords are completely removed, the system focuses on semantically meaningful terms to enhance discriminative power and retrieval accuracy.

Formula 2: Neighbor Retrieval (Sentence→Sentence)

Purpose: Evaluate the contextual coherence between the main retrieved sentence and its neighboring sentences (previous or next). This process ensures that context expansion maintains both semantic continuity and lexical consistency

Word Weight Assignment: Unlike Formula 1's query-level dynamic balancing, Formula 2 now applies **token-level dynamic weighting** based on word frequency to better capture thematic relevance between sentences. :

$$(\lambda_{\text{lex}}, \lambda_{\text{sem}}) = \begin{cases} (0.30, 0.70) & \text{if length_ratio} \geq 1.5 \text{ (much longer)} \\ (0.40, 0.60) & \text{if } 1.2 \leq \text{length_ratio} < 1.5 \text{ (longer)} \\ (0.50, 0.50) & \text{if } 0.8 \leq \text{length_ratio} < 1.2 \text{ (similar)} \\ (0.60, 0.40) & \text{if } 0.5 \leq \text{length_ratio} < 0.8 \text{ (shorter)} \\ (0.70, 0.30) & \text{if length_ratio} < 0.5 \text{ (much shorter)} \end{cases}$$

Final Score Computation (Formula 2):

$$\text{neighbor_score} = (\lambda_{\text{sem}} \times \text{neighbor_semantic}) + (\lambda_{\text{lex}} \times \text{neighbor_lexical})$$

where:

- **neighbor_score** — represents the overall similarity score between the main retrieved sentence and its neighboring sentence.
- λ_{sem} — is the weight assigned to the semantic similarity component.
- **neighbor_semantic** — is the semantic similarity score between the main and neighboring sentence, computed using cosine similarity between their sentence embeddings.
- λ_{lex} — is the weight assigned to the lexical similarity component.
- **neighbor_lexical** — is the lexical similarity score between the main and neighboring sentence, calculated using the Jaccard coefficient as defined in Section 3.5.4.

Relevance Threshold: Neighbors are included as context only if:

$$\text{if neighbor_score} \geq 0.40$$

Rationale: Context expansion relies on maintaining strong semantic coherence to preserve the thematic flow between adjacent sentences. Instead of fixed proportional weighting, Formula 2 uses dynamic frequency-based weighting to automatically emphasize rare and contextually significant words. The Jaccard coefficient remains integral for measuring lexical continuity, while

adaptive word weights ensure that meaningful terms contribute more to the overall score than high-frequency or stopword terms. This approach ensures that neighboring sentences are selected based on conceptual relevance rather than mere word overlap, with the higher relevance threshold (0.40 vs. 0.30) filtering out loosely related or tangential content.

3.6.7 Formula Application Example

To illustrate how the dual formulas operate in practice, consider the query "Edukasyon bilang pag-asa ni Ibarra" (Education as Ibarra's hope):

Step 1: Query Validation

Linguistic Validation:

- Total words: 5 (Edukasyon, bilang, pag-asa, ni, Ibarra)
- Content words: 3 (Edukasyon, pag-asa, Ibarra)
- Stopwords: 2 (bilang, ni)
- Stopword ratio: 40.0%
- Valid word ratio: 100% ✓ PASS

Semantic Validation:

- Domain alignment: $\max_{p \in C} \text{CosineSim}(\mathbf{e}_q, \mathbf{e}_p) = 0.499$ (> 0.30 threshold) ✓ PASS
- Theme proximity: 0.794 (> 0.35 threshold) ✓ PASS
- Query proceeds to retrieval

Step 2: Formula 1 Weight Calculation

Since $L = 5$ (counting content words) and $r_{\text{stop}} = 0.40 < 1.0$:

$$(\lambda_{\text{sem}}, \lambda_{\text{lex}}) = (0.75, 0.25)$$

No stopwords adjustment needed (stopword ratio below 1.0 threshold).

Step 3: Main Retrieval Scoring

For the top result (Chapter 19, Sentence 5):

Passage text: "Nagtanong si Ibarra kung ano ang kinahinatnan ng pagtulong ng ama sa mga mahihirap. Sinabi ng guro na nakatulong si Don Rafael sa pagpapaunlad ng edukasyon sa kanilang bayan."

Semantic similarity: $\text{SemanticSim}(q, p) = 0.499$ (49.9%)

- Computed via cosine similarity between XLM-RoBERTa embeddings
- High similarity captures conceptual alignment (Ibarra + education + help)

Lexical overlap:

- Matched content words: {edukasyon, ibarra}
- Semantic weights: $w_{\text{edukasyon}} = 0.7, w_{\text{pag-asa}} = 0.7, w_{\text{ibarra}} = 1.0$
- Total query weight: $0.7 + 0.7 + 1.0 = 2.4$
- Matched weight: $0.7 + 1.0 = 1.7$ (edukasyon and ibarra found in passage)
- $\text{LexicalOverlap}(q, p) = \frac{1.7}{2.4} = 0.708$ (70.8%)

Formula 1 final score:

$$\text{Score}_{\text{main}} = (0.75 \times 0.499) + (0.25 \times 0.708) - 0 = 0.374 + 0.177 = 0.551$$

Final score: 55.1% (shown as "Final" in output)

Step 4: Context Expansion with Formula 2

For the previous sentence (Sentence 4):

Main sentence (S5): "Sinabi ng guro na nakatulong si Don Rafael sa pagpapaunlad ng edukasyon sa kanilang bayan."

Neighbor sentence (S4): "Ayon sa guro ay wala dapat itong ipagpasalamat sapagkat malaki ang utang na loob nito kay Don Rafael dahil isa ito sa mga nabigyan ng tulong nung ito'y nabubuhay pa."

Semantic similarity (sentence-to-sentence): $\text{SemanticSim}(m, n) = 0.800$
(80.0%)

Lexical similarity (Jaccard):

- Main words: {sinabi, guro, nakatulong, don, rafael, pagpapaunlad, edukasyon, kanilang, bayan}
- Neighbor words: {ayon, guro, wala, dapat, itong, ipagpasalamat, sapagkat, malaki, utang, loob, nito, kay, don, rafael, dahil, isa, ito, mga, nabigyan, tulong, nung, nabubuhay, pa}
- Intersection: {guro, don, rafael} = 3 words (after stopword filtering)
- Union: 26 unique words

- $\text{LexicalSim}_{\text{neighbor}} = \frac{3}{26} = 0.182 \text{ (18.2\%)}$

Formula 2 neighbor score:

$$\text{Score}_{\text{neighbor}} = (0.65 \times 0.800) + (0.35 \times 0.182) = 0.520 + 0.064 = 0.584$$

Neighbor score: 58.4% (exceeds 0.40 threshold → included as context)

The system continues this process for all adjacent sentences, stopping when:

- Neighbor score falls below 0.40 threshold
- Maximum distance of 5 sentences reached
- Chapter boundary encountered

Step 5: Result Display

The output shows:

▮ MAIN RETRIEVAL (Query→Corpus): $\lambda_{\text{lex}}=0.25$ | $\lambda_{\text{sem}}=0.75$

▮ NEIGHBOR RETRIEVAL (Sentence→Sentence): $\lambda_{\text{lex}}=0.35$ | $\lambda_{\text{sem}}=0.65$

Result 1:

Semantic: 49.9% | Lexical: 70.8% | Final: 55.2%

Neighbor Similarity Metrics:

Previous S4: Semantic: 80.0% | Lexical: 18.2% | Combined: 58.4%

▮ This example demonstrates how:

- Query passed both linguistic and semantic validation
- Formula 1 dynamically balanced semantic (75%) and lexical (25%) for the 5-word query
- Formula 2 used dynamic weights (65% semantic, 35% lexical) to assess context coherence
- The neighbor achieved high relevance (58.4%) despite low lexical overlap (18.2%) because of strong semantic coherence (80.0%)
- Jaccard coefficient appropriately measured vocabulary continuity between adjacent sentences

3.6.8 Query Suggestion Component

To assist users in formulating effective queries, particularly students unfamiliar with thematic terminology, the system implements intelligent query suggestions for incomplete inputs.

3.6.8.1 Single-Word Theme Input

When a user types a single thematic term (e.g., "edukasyon"), the system:

Theme Detection and Character Association

- **Theme Recognition:** Identifies the input as a thematic keyword using:
 - Comparison against predefined theme vocabulary (Section 3.3.3)
 - Semantic similarity to theme embeddings (threshold ≥ 0.60)
- **Character Co-occurrence Mining:** Extracts characters most frequently associated with the theme across the corpus using:
 - Named Entity Recognition (NER) via spaCy to identify PROP entities

- Co-occurrence frequency within same passages
- Semantic proximity in embedding space
- **Relational Marker Selection:** Determines appropriate possessive/relational marker based on grammatical context:
 - "ni" (personal possessive): For individual characters
 - "ng" (general possessive): For groups, institutions, or abstract entities
 - "sa" (locational/indirect): For place-based or indirect relationships
- **Template Construction:** Generates suggestions using pattern:

□[THEME] + [MARKER] + [CHARACTER/ENTITY]

□Example: Suggestions for "edukasyon"

Input: "edukasyon"

System Process:

- Detects "edukasyon" as theme (matches theme: "Edukasyon")
- Extracts co-occurring entities from corpus:
 - Characters: {Ibarra, Don Rafael, mga estudyante, guro}
 - Institutions: {paaralan, mga prayle, San Diego}
- Computes co-occurrence frequencies and semantic similarities
- Selects appropriate grammatical markers

Generated Suggestions (ranked by relevance):

1. "edukasyon ni Ibarra" (ni - personal possessive for main protagonist)
2. "edukasyon ni Don Rafael" (ni - personal possessive for Ibarra's father)
3. "edukasyon ng mga estudyante" (ng - general possessive for group)
4. "edukasyon sa San Diego" (sa - locational marker for place)
5. "edukasyon ng mga prayle" (ng - possessive for institution/group)

3.6.8.2 Single-Word Character Input

When a user types a single character name (e.g., "Basilio"), the system:

Character Detection and Theme Association

- **Entity Recognition:** Identifies the input as a character name using:
 - Part-of-speech tagging (PROPN) via spaCy
 - Matching against character registry extracted from corpus
- **Thematic Action Extraction:** Extracts themes and actions associated with the character using:
 - spaCy dependency parsing to find verbs and nouns co-occurring with the character
 - Semantic clustering of extracted terms into thematic categories
- **Grammatical Marker Application:** Always uses "ni" for personal character possession
- **Template Construction:** Generates suggestions using pattern:

□[THEME/ACTION] + "ni" + [CHARACTER]

Example: Suggestions for "Basilio"

Input: "Basilio"

System Process:

- Detects "Basilio" as character entity (POS: PROPEN)
- Extracts co-occurring thematic terms: {kabayanihan, pag-aaral, buhay, paghihirap, pamilya}
- Ranks by embedding similarity to character's narrative context
- Applies "ni" marker consistently

Generated Suggestions (ordered by similarity score):

1. "kabayanihan ni Basilio"
2. "pag-aaral ni Basilio"
3. "buhay ni Basilio sa San Diego"
4. "paghihirap ng pamilya ni Basilio" (note: "ng pamilya" for group possession)
5. "pagtulong ni Basilio"

3.6.8.3 Entity-Based Chapter Location with Domain-Adaptive Semantic Grounding

When a user enters a standalone entity without thematic context (e.g., "Sisa", "San Diego", "paaralan"), the system maps its location across both *Noli Me Tangere* and *El Filibusterismo* using domain-adaptive semantic validation.

Entity Detection and Validation

The system uses spaCy to identify PROPEN (proper nouns) and NOUN (common nouns) as entity queries. Before mapping, entities undergo semantic validation by computing maximum cosine similarity between entity embedding and corpus passages. Validation requires semantic alignment threshold of 0.35, at least 2 corpus occurrences, and 5 surrounding content words. Invalid entities receive rejection messages with diagnostic information.

Chapter Mapping Process

Prominence Calculation: For each chapter containing the entity, prominence is calculated by combining mention count (40% weight) with average semantic relevance (60% weight). Prominence classifies as Primary when score is 0.70 or above, Secondary for scores between 0.45-0.69, or mentioned for scores between 0.20-0.44.

Cross-Novel Mapping: The system identifies which novel(s) contain the entity, lists all chapters where it appears, determines the narrative span (first to last chapter), and identifies the peak prominence location.

Contextual Information

For each chapter location, the system establishes thematic connections for themes scoring threshold of 0.45 or above. Themes are derived from predefined literary sources including colonialism (kolonyalismo), education (edukasyon), religion (relihiyon), social inequality (hindi pagkakapantay-pantay sa lipunan), justice (katarungan), revolution (rebolusyon), violence versus reform (karahasan laban sa reporma), and greed (kasakiman).

Display Format

Output shows entity name with semantic alignment score, separate sections for Noli Me Tangere and El Filibusterismo indicating presence or absence, chapter listings with prominence levels and thematic connections, and related search suggestions.

Application Examples

Example 1: Character Search - "Sisa"

ENTITY: Sisa (Character)

Semantic Alignment: 78.0% ✓ Validated

 NOLI ME TANGERE

Narrative Span: Chapters 6-20

Total Mentions: 47

Chapter 6: Si Kapitan Tiyago [Secondary - 0.64]

- Themes: Pamilya (Family), Kahirapan (Poverty)

Chapter 15: Ang mga Sakristan [Primary - 0.72]

- Themes: Katiwalian (Corruption), Karahasan (Violence)

Chapter 17: Ang Mamatay na Ina [Primary - 0.89]

- Themes: Pagdurusa (Suffering), Pang-aapi (Oppression)

Chapter 20: Si Sisa [Primary - 0.85]

- Themes: Karahasan sa Pamilya (Family Violence)

 EL FILIBUSTERISMO

X Does not appear

Related Searches: "Pagdurusa ni Sisa" | "Karahasan sa pamilya"

Example 2: Concept Search - "edukasyon"

ENTITY: edukasyon (Concept)

Semantic Alignment: 85.0% ✓ Validated

 NOLI ME TANGERE

Narrative Span: Chapters 7-55

Total Mentions: 89

Chapter 7: Ang Hapunan [Secondary - 0.68]

- Themes: Kolonyalismo (Colonialism), Edukasyon (Education)

Chapter 18: Ang mga Batang Tao [Primary - 0.92]

- Themes: Edukasyon (Education), Pag-asa (Hope)

Chapter 23: Ang Pagtitipon [Primary - 0.87]

- Themes: Relihiyon (Religion), Kapangyarihan ng Simbahan

EL FILIBUSTERISMO

Narrative Span: Chapters 3-14

Total Mentions: 34

Chapter 3: Ang mga Makapangyarihan [Secondary - 0.63]

- Themes: Katiwalian (Corruption), Sistema ng Pamahalaan

Chapter 10: Yaman at Karalitaan [Mentioned - 0.42]

- Themes: Hindi Pagkakapantay-pantay (Social Inequality)

Related Searches: "Paaralan ni Ibarra" | "Kolonyalismo sa edukasyon"

System Integration

This mapping component extends the query suggestion architecture by applying domain-adaptive validation with semantic alignment threshold of 0.35. It uses the main retrieval formula for entity-passage matching while leveraging thematic classification with similarity threshold of 0.45 to enrich contextual information. The implementation provides systematic access to narrative element locations, addressing student difficulties in tracking entities across both novels by showing where characters, places, objects, and concepts appear with their associated themes from established literary sources.

3.7 Thematic Classification

Retrieved passages undergo thematic classification to identify which predefined themes they exemplify, addressing the research objectives.

Thematic Classification and Analysis Process

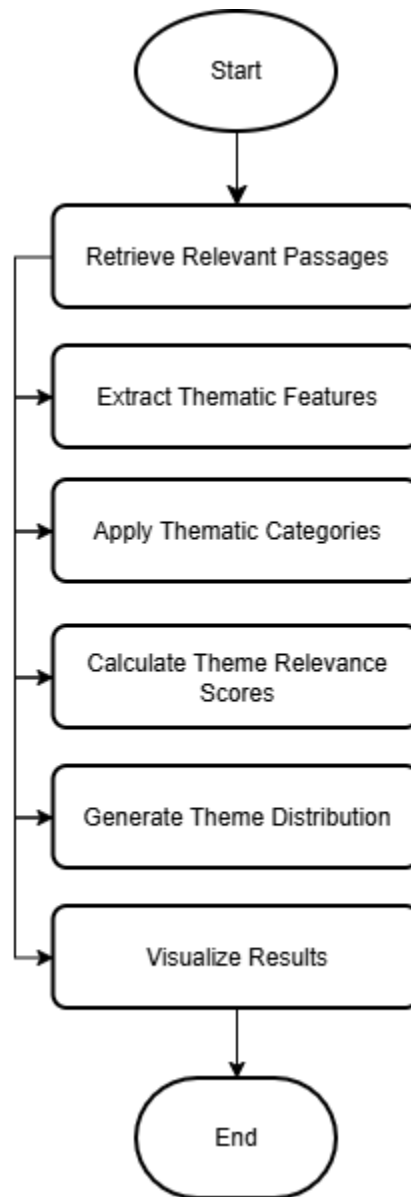


Figure 10: Thematic Classification and Analysis Process

Theme Matching Process:

Theme Encoding: Each predefined theme is represented as concatenated text:

$$\text{theme_text} = \text{Tagalog Title} + \text{Meaning}$$

Example: "Edukasyon Sa mga mata ni Ibarra, ang kaalaman ay sagisag ng pag-asa, habang sa mga prayle, ito'y panganib sa kanilang kapangyarihan"

Theme Embedding: Generate embeddings for all themes:

$$\mathbf{e}_{\text{theme}} = \text{XLM-RoBERTa}(\text{theme_text})$$

Passage-Theme Similarity: For each retrieved passage, compute similarity to all themes:

$$s_{\text{theme}}(p, t) = \text{CosineSim}(\mathbf{e}_p, \mathbf{e}_t)$$

Theme Assignment: Themes with $s_{\text{theme}} \geq 0.45$ are assigned to the passage. The highest-scoring theme becomes the "primary theme."

Thematic Coverage Metrics:

Thematic Coverage: Proportion of retrieved passages with at least one assigned theme

Average Theme Confidence: Mean similarity score across assigned themes

Classification Type: Results classified as "thematic analysis" if coverage $\geq 30\%$ and average confidence ≥ 0.45 , otherwise "semantic search"

Example from "Edukasyon bilang pag-asa ni Ibarra" query:

For Result 1 (Chapter 19, Sentence 5):

- Passage embedding computed from retrieved sentence
- Compared against all theme embeddings
- Primary theme: "Edukasyon" with confidence 79.4%
- This indicates strong thematic alignment between retrieved passage and education theme

This thematic layer enables analysis beyond passage retrieval, identifying which literary themes are present and how they distribute across the novels.

3.8 Evaluation Methodology

The evaluation strategy combines quantitative performance metrics with qualitative expert assessment to comprehensively evaluate the dual-formula system's effectiveness, directly addressing the research objectives.

Evaluation Metrics and Framework

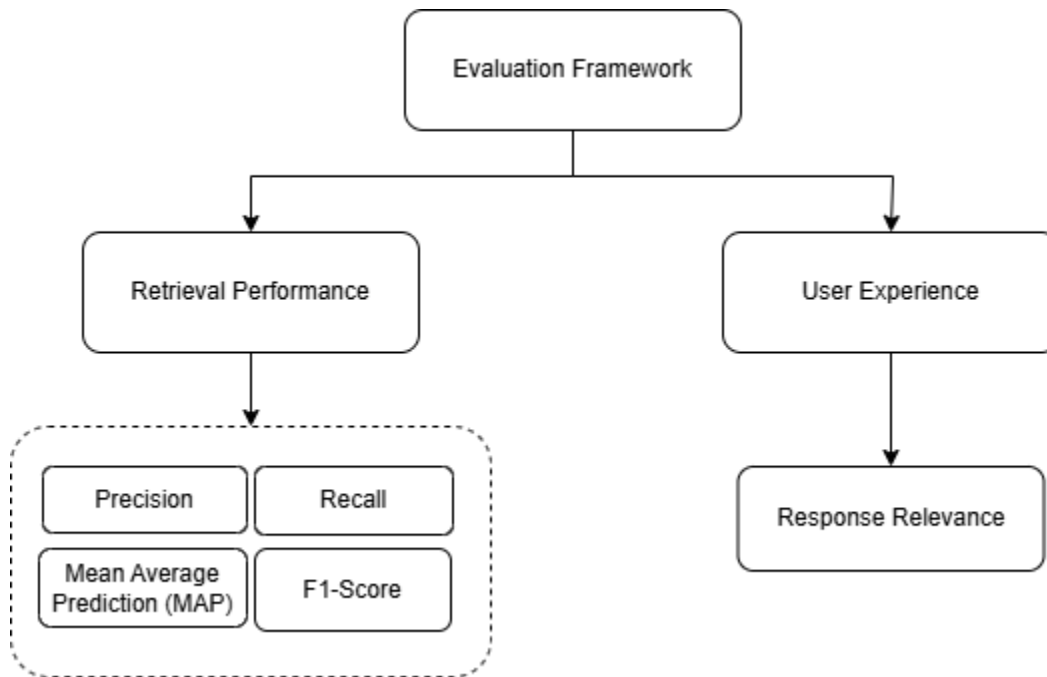


Figure 11: Evaluation Metrics and Framework

3.8.1 Baseline Comparison

System performance is evaluated against a keyword-only baseline that uses only the lexical overlap component without semantic understanding:

$$\text{Score}_{\text{baseline}}(q, p) = \text{LexicalOverlap}(q, p)$$

This baseline represents traditional Ctrl+F keyword search approaches criticized in the problem statement. The comparison demonstrates the incremental value provided by:

- Semantic understanding via XLM-RoBERTa embeddings
- Hybrid integration via Formula 1

- Context expansion via Formula 2
- Domain-adaptive semantic validation

3.8.2 Quantitative Evaluation Metrics

Precision and Recall: For each theme query, a relevance judgment set is created by manually annotating passages as relevant or non-relevant to that theme.

$$\text{Precision} = \frac{|\text{Retrieved} \cap \text{Relevant}|}{|\text{Retrieved}|}$$

$$\text{Recall} = \frac{|\text{Retrieved} \cap \text{Relevant}|}{|\text{Relevant}|}$$

F1-Score: The harmonic mean of precision and recall:

$$\text{F1} = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

Mean Average Precision (MAP): To account for ranking quality:

$$\text{MAP} = \frac{1}{|Q|} \sum_{q=1}^{|Q|} \frac{1}{m_q} \sum_{k=1}^{m_q} \text{Precision@}k \cdot \text{rel}(k)$$

where $|Q|$ is the number of queries, m_q is the number of relevant documents for query q , and $\text{rel}(k)$ is 1 if the document at rank k is relevant, 0 otherwise.

Domain Validation Metrics:

True Negative Rate (Specificity): For known-negative queries (queries about events/characters not in the novels):

$$\text{TNR} = \frac{\text{Correct Rejections}}{\text{Total Negative Queries}}$$

False Positive Rate: Proportion of invalid queries that incorrectly return results:

$$\text{FPR} = \frac{\text{Incorrect Retrievals}}{\text{Total Negative Queries}} = 1 - \text{TNR}$$

Precision at Valid Queries: For queries that should return results:

$$\text{Precision}_{\text{valid}} = \frac{\text{Correct Retrievals}}{\text{Total Valid Queries}}$$

Component Contribution Analysis (Ablation Studies):

Compare performance across:

1. **Semantic-only:** $\lambda_{\text{sem}} = 1, \lambda_{\text{lex}} = 0$ (Formula 1)
2. **Lexical-only:** $\lambda_{\text{sem}} = 0, \lambda_{\text{lex}} = 1$ (Formula 1)
3. **Formula 1 with static weights:** $\lambda_{\text{sem}} = 0.5, \lambda_{\text{lex}} = 0.5$ (no dynamic adjustment)
4. **Formula 1 without length penalty:** $\gamma = 0$
5. **Context expansion using Formula 1 instead of Formula 2**
6. **Without domain validation:** All queries proceed to retrieval regardless of semantic grounding
7. **Full dual-formula model:** Dynamic Formula 1 + Dynamic Formula 2 + Domain validation

This quantifies each component's individual contribution to overall performance and validates the effectiveness of using two specialized formulas with domain-adaptive validation.

3.8.3 Qualitative Expert Review

Two to three literature educators or scholars with expertise in Rizal's works evaluate retrieval quality.

Evaluation Protocol:

Query Selection: Experts presented with:

- 5 valid thematic queries (3 from *Noli Me Tangere*, 2 from *El Filibusterismo*)
- 3 known-negative queries (about events/characters not in the novels)

Result Presentation: For each valid theme, top 10 retrieved passages from both hybrid and baseline systems shown in random order without system identification.

Relevance Rating: Experts rate each passage on 5-point Likert scale:

- 1 = Not Relevant
- 2 = Slightly Relevant
- 3 = Moderately Relevant
- 4 = Highly Relevant
- 5 = Perfectly Relevant

Rejection Assessment: For known-negative queries, experts evaluate:

- Appropriateness of rejection (should the system have been rejected?)
- Quality of rejection explanation
- Whether any returned results (if system failed to reject) are actually relevant

Qualitative Feedback: Experts provide written comments on:

- Thematic alignment quality
- Context coherence (relevance of neighbor sentences)
- Effectiveness of domain validation in preventing false results
- Overall system utility for literary analysis

Analysis: Expert ratings aggregated to compute mean relevance scores. Inter-rater reliability assessed using Krippendorff's alpha ($\alpha \geq 0.67$ indicates acceptable agreement). Qualitative comments analyzed thematically to identify system strengths and weaknesses, particularly regarding:

- Formula 2's effectiveness in selecting coherent context
- Domain validation's success in preventing semantic hallucination
- Balance between rejection strictness and retrieval coverage

3.8.4 Thematic Distribution and Co-occurrence Analysis

To address the research objectives, the study analyzes how themes distribute across and co-occur within both novels.

Distribution Analysis:

For each theme, identify all passages scoring above relevance threshold (0.45 theme similarity).

Count theme frequency per chapter.

Visualize using:

- Bar charts showing theme frequency per novel
- Line graphs showing theme concentration across narrative progression
- Heatmaps showing chapter-level theme prevalence

Co-occurrence Analysis:

Identify passages where multiple themes simultaneously score above threshold.

Construct co-occurrence matrix: C_{ij} = count of passages containing both theme i and theme j

Visualize using:

- Heatmaps of pairwise theme co-occurrence frequencies
- Network graphs (nodes = themes, edges = co-occurrence strength)
- Chord diagrams for multi-theme relationships

3.8.5 Query Complexity Testing

To comprehensively evaluate the system's capability to handle diverse user input patterns, the evaluation includes queries of varying complexity levels representative of actual student usage:

System Testing and Validation Workflow

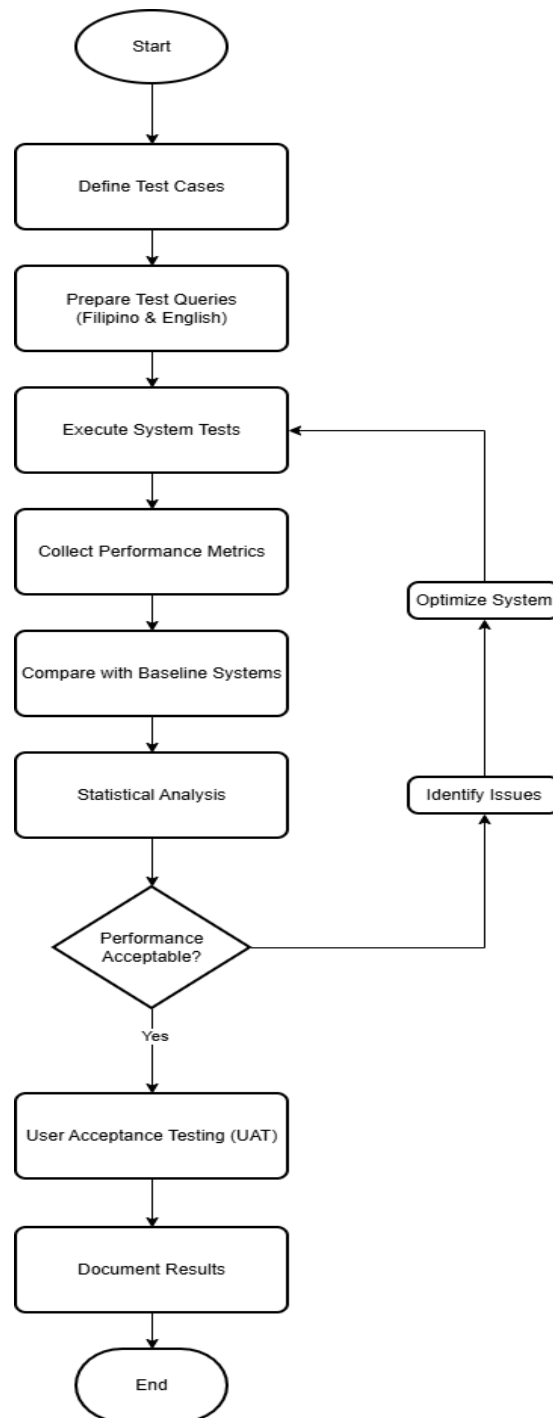


Figure 12: System Testing and Validation Workflow

Simple Character-Action Queries (Basic Relational Patterns)

These represent typical high-school level searches combining a character with thematic verbs or nouns:

- "edukasyon ni Ibarra"
- "pagmamahal ni Maria Clara"
- "kamatayan ni Basilio"
- "kabaitan ni Padre Salvi"

These inputs test whether the system understands basic relational patterns commonly used by students in literary analysis.

Multi-Concept & Multi-Relation Queries (Complex Semantic Patterns)

These queries involve multiple themes, characters, and emotional or moral relationships requiring deeper semantic validation:

- "kabaitan ni Don Rafael sa mga dukha"
- "galit at kasalanan ni Padre Damaso kay Ibarra"
- "pagdurusa ni Sisa dahil sa kawalan ng hustisya"
- "pang-aapi ng mga prayle sa pamilya ni Ibarra"

These complex samples test the system's ability to detect:

- Semantic contradictions (e.g., "kabaitan ni Padre Damaso")

- Multi-theme coherence across character relationships
- Cross-character relational dynamics
- Correctness based on canonical novel context

The domain-adaptive semantic validation (Section 3.2.2) is specifically designed to reject queries containing semantic contradictions (e.g., ascribing virtue to antagonistic characters) while accepting valid multi-concept queries that reflect actual narrative relationships.

Statistical Testing:

Test whether theme co-occurrence exceeds random chance:

$$\chi^2 = \sum \frac{(O_{ij} - E_{ij})^2}{E_{ij}}$$

where O_{ij} is observed co-occurrence and E_{ij} is expected co-occurrence under independence assumption.

This analysis reveals thematic patterns potentially obscured in traditional close-reading, providing empirical evidence for claims about thematic interconnections in Rizal's novels.

3.9 Implementation Details

3.9.1 Development Environment and Tools

Programming Language: Python 3.9+

Key Libraries:

- **sentence-transformers (4.0+):** XLM-RoBERTa model loading and inference
- **scikit-learn (1.3+):** Cosine similarity computation, evaluation metrics
- **pandas (2.0+):** Data manipulation and passage metadata management
- **numpy (1.24+):** Numerical computations for scoring functions
- **stopwordsiso (0.6+):** Official Tagalog stopwords from ISO 639
- **wordfreq (3.0+):** Word frequency validation for Filipino
- **rich (13.0+):** Terminal-based result visualization with dual-formula metrics display

Hardware: Development conducted on standard CPU-based hardware (8GB RAM minimum). XLM-RoBERTa base model inference is computationally feasible without GPU acceleration for the corpus size used.

3.9.2 Data Storage Format

Chapters CSV Structure:

- **chapter_number:** Integer chapter identifier
- **chapter_title:** String chapter title
- **sentence_number:** Integer sentence identifier within chapter
- **sentence_text:** String passage content

Themes CSV Structure:

- **Tagalog Title:** String theme name in Filipino
- **Meaning:** String theme description/explanation
- **Novel:** String identifier (Noli/Fili)

3.9.3 System Workflow

Initialization Phase (Offline):

1. Load **noli_chapters.csv**, **elfili_chapters.csv**, **noli_themes.csv**, **elfili_themes.csv**
2. Initialize XLM-RoBERTa model via SentenceTransformer wrapper
3. Load official Tagalog stopwords from stopwords-iso
4. Generate and cache embeddings for all passages (combined chapter title + sentence text) and themes

5. Compute domain centroid and semantic boundaries from corpus embeddings
6. Compute sentence word counts for length penalty
7. Build corpus vocabulary from passages and themes for lexical grounding validation

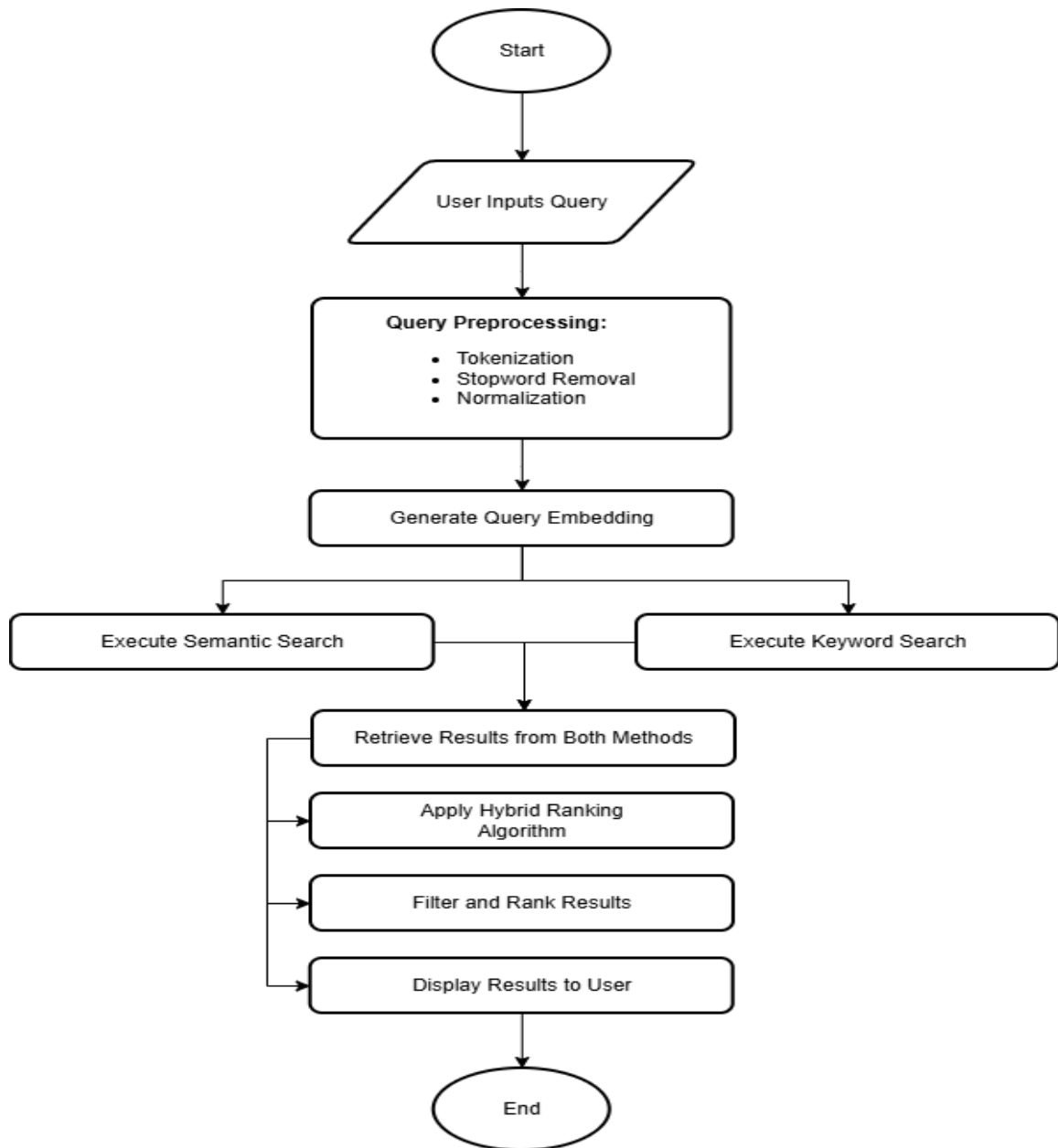
Query Phase (Online):

Figure 13: Query Processing Execution Workflow

Step 1: Receive and Validate Query

- Receive thematic query from user
- **Phase 1 Validation (Linguistic):**
 - Check for valid Filipino words using wordfreq
 - Verify basic lexical presence in corpus vocabulary
 - Compute stopwords ratio
 - If fails: Return linguistic validation error

Step 2: Domain-Adaptive Semantic Validation

- Generate query embedding using XLM-RoBERTa
- Compute domain alignment: $\max_{p \in C} \text{CosineSim}(\mathbf{e}_q, \mathbf{e}_p)$
- Compute theme proximity: $\max_{t \in T} \text{CosineSim}(\mathbf{e}_q, \mathbf{e}_t)$
- Apply rejection criteria:
 - If $\text{DomainAlignment} < 0.30 \wedge \text{ThemeProximity} < 0.35$: **REJECT**
 - Return: "No thematically or semantically relevant passage found."
- If passes: Proceed to retrieval

Step 3: Query Analysis

- Extract words using regex pattern (preserving hyphens)
- Identify stopwords using stopwords-iso
- Compute word frequencies
- Assign semantic weights to each word
- Calculate query length and stopword ratio

Step 4: Execute Formula 1 (Main Retrieval)

- Compute semantic similarities for all passages (filter by 0.20 threshold)
- Compute weighted lexical overlap scores
- Calculate dynamic weights based on query length and stopword ratio
- Compute multi-component final scores using Formula 1
- Sort by final score
- Apply ranking constraints:
 - Maximum 3 passages per chapter (diversity)
 - Mark used passages for deduplication

Step 5: Execute Formula 2 (Context Expansion)

- For each top-k passage, identify adjacent sentences
- Compute semantic similarity (sentence-to-sentence)
- Compute lexical similarity using Jaccard coefficient
- Calculate neighbor scores using Formula 2 weights
- Include neighbors exceeding 0.40 threshold

- Stop expansion at irrelevant sentence or chapter boundary
- Mark context sentences as used

Step 6: Perform Thematic Classification

- Compare passages to theme embeddings
- Assign themes exceeding 0.45 similarity threshold
- Identify primary themes

Step 7: Format and Display Results

- Show dual-formula metrics (Formula 1 weights, Formula 2 scores)
- Display neighbor similarity breakdown with Jaccard lexical scores
- Present thematic analysis
- For rejected queries: Show diagnostic information (domain alignment, theme proximity, rejection reason)

Evaluation Phase:

1. Execute retrieval for all test themes (valid and known-negative)
2. Record domain validation decisions (accept/reject)
3. Compare results against manual relevance judgments
4. Compute precision, recall, F1, MAP for each configuration
5. Compute TNR, FPR for domain validation
6. Conduct ablation studies (semantic-only, lexical-only, single formula, without validation, etc.)
7. Facilitate expert review sessions with randomized result presentation
8. Generate distribution and co-occurrence visualizations

9. Perform statistical testing on theme patterns

3.9.4 System Parameters

Domain Validation:

- Domain alignment threshold: 0.30
- Theme proximity threshold: 0.35
- Combined rejection rule: Both thresholds must fail

Main Retrieval (Formula 1):

- Minimum semantic threshold: 0.20
- Short sentence threshold: 5 words
- Short sentence penalty: 0.08
- High stopword ratio threshold: 1.0
- Length penalty coefficient (γ): 1.0

Neighbor Retrieval (Formula 2):

- Semantic weight ($\lambda_{\text{sem}}^{\text{neighbor}}$): Dynamic, adjusts based on contextual relevance
- Lexical weight ($\lambda_{\text{lex}}^{\text{neighbor}}$): Dynamic, adjusts based on corpus frequency distribution
- Lexical similarity metric: Jaccard coefficient
- Neighbor relevance threshold: 0.40
- No limit on context expansion in either direction

Thematic Classification:

- Theme assignment threshold: 0.45
- Thematic coverage threshold: 30%
- Average confidence threshold: 0.45

General:

- Valid Filipino word frequency threshold: 1×10^{-8}
- Valid word ratio requirement: 100%
- Maximum passages per chapter: 3
- Top-k results returned: 9

3.10 Ethical Considerations

Data Sources: The corpus consists of publicly available educational summaries of classic literature in the public domain. No copyrighted material is reproduced beyond fair use for academic research purposes.

Attribution: All theoretical frameworks, methodologies, and external sources are properly cited following academic standards.

Expert Review Ethics: Participating literature educators/scholars provide informed consent and are informed that evaluations are for research purposes. No personally identifiable information is collected beyond expertise credentials. Participation is voluntary and can be withdrawn at any time.

Reproducibility: The methodology is documented in sufficient detail to enable replication. Code implementation will be made available upon reasonable request to support future research, subject to ethical review.

Cultural Sensitivity: The study recognizes José Rizal's novels as culturally significant works in Philippine national identity. The research aims to enhance accessibility and understanding of these works rather than replace human literary scholarship or impose external interpretive frameworks.

Bias Mitigation: The dual-formula hybrid system with domain-adaptive validation is designed to minimize algorithmic bias by:

- Using multilingual models pre-trained on diverse corpora
- Incorporating multiple relevance signals (semantic + lexical) to reduce dependence on any single representation
- Implementing domain validation to prevent semantic hallucination and false results
- Validating results through expert review by Filipino literature specialists
- Transparently documenting system limitations and delimitations
- Making dual-formula scoring process interpretable and explainable

Query Rejection Transparency: The domain-adaptive validation mechanism explicitly informs users when queries are rejected and why, preventing confusion and maintaining trust in system outputs.

3.11 Limitations and Delimitations

3.11.1 Methodological Limitations

Use of Summary Versions: While summaries preserve core thematic content, they necessarily omit narrative details, character development nuances, and linguistic subtleties present in full texts. Retrieval results reflect themes as represented in summaries rather than original novels. However, these summaries represent actual educational materials used by Filipino students, increasing practical relevance.

No Fine-tuning: The XLM-RoBERTa model is used without fine-tuning on Filipino literary texts due to lack of annotated training data. While cross-lingual transfer enables effective performance (Imperial, 2021), fine-tuning could potentially improve semantic understanding of Filipino literary language and archaic expressions.

Manual Relevance Judgments: Quantitative evaluation requires manual annotation of passage relevance, which is inherently subjective despite use of established theme definitions. This limitation is partially mitigated through:

- Use of multiple annotators with inter-rater reliability assessment
- Grounding annotations in established literary sources (GradeSaver, LitCharts)
- Supplementing quantitative metrics with qualitative expert review

Limited Corpus Scope: Evaluation focuses on two novels only. Generalization to other Filipino literary works (e.g., *Ibong Adarna*, *Florante at Laura*) requires additional validation. The dual-formula methodology is designed to be adaptable, but performance may vary with different writing styles, time periods, and linguistic registers.

Context Window Limitations: XLM-RoBERTa has a 512-token maximum input length. For longer passages (rare in summaries but possible), this may affect embedding quality. However, summary sentences typically remain well within limits.

Formula Parameter Selection: The weighting behavior and thresholds in Formula 2 were determined empirically through pilot testing to optimize sentence-to-sentence coherence. Instead of fixed semantic-lexical ratios, the system employs dynamic frequency-based weights that adjust according to corpus characteristics. Different literary datasets or retrieval objectives may still benefit from tuning frequency thresholds or relevance cutoffs to achieve optimal contextual performance.

Domain Validation Threshold Sensitivity: The domain alignment threshold of 0.30 was calibrated using pilot testing with known-positive and known-negative queries. While this threshold effectively prevents semantic hallucination in tested scenarios, edge cases near the boundary may produce false rejections (blocking valid queries) or false acceptances (allowing weakly related queries). The dual-threshold system (0.30 for domain alignment, 0.35 for theme proximity) provides additional robustness but requires periodic recalibration when corpus content changes significantly.

3.11.2 Delimitations

Predefined Themes Only: The system retrieves passages for predefined themes from established literary sources rather than discovering themes automatically through unsupervised methods (e.g., LDA, NMF). This design choice:

- Focuses research on retrieval effectiveness rather than theme extraction
- Ensures evaluation against validated thematic frameworks
- Aligns with educational use cases where teachers query specific curriculum themes

- Acknowledges that automated theme discovery for Filipino literature represents a separate, substantial research question

No Large-Scale User Studies: Evaluation relies on expert review rather than extensive user testing with students or teachers, acknowledging resource constraints of a thesis project. This delimitation means:

- Technical performance is validated but pedagogical effectiveness in classroom settings is not empirically measured
- User experience and usability aspects receive limited evaluation
- Future work should validate educational effectiveness through classroom deployment and longitudinal studies

Single Model Architecture: The study uses XLM-RoBERTa exclusively without comparing against other multilingual models (e.g., mBERT, mT5, LaBSE). This choice was made because:

- Literature review (Chapter 2) established XLM-RoBERTa as optimal for Filipino code-switched text (Salve & Tubil, 2025; Cosme & De Leon, 2024)
- Resource constraints limit comprehensive model comparison
- The research focuses on dual-formula hybrid architecture design with domain-adaptive validation rather than model selection

Sentence-Level Granularity: Passages are defined as individual sentences rather than multi-sentence paragraphs or variable-length discourse units. This was chosen for:

- Clear semantic boundaries
- Alignment with summary structure

- Computational efficiency
- Precise theme localization

However, some themes may span multiple sentences, requiring context expansion to capture full thematic expression—which Formula 2 addresses.

No Temporal Analysis: The study does not analyze how themes evolve chronologically within each novel's narrative progression. While distribution visualizations show where themes appear, detailed temporal/narrative arc analysis is beyond the current scope.

Binary Formula Design: The system uses exactly two formulas (main and neighbor) rather than exploring a continuous spectrum of weighting schemes or additional specialized formulas for other retrieval subtasks.

Summary Corpus Only: The domain-adaptive semantic validation is calibrated specifically for the summary corpus semantic space. Queries about content present in the original novels but omitted from summaries may be incorrectly rejected as semantically misaligned. This is an inherent trade-off of using summary versions rather than full texts.

3.11.3 Validation Boundaries

Thematic Classification Threshold: The 0.45 cosine similarity threshold for theme assignment was determined empirically through pilot testing. This threshold balances precision (avoiding false theme assignments) with recall (capturing relevant themes). Different thresholds may be appropriate for different use cases or literary corpora.

Neighbor Threshold: The 0.40 combined score threshold for Formula 2 was set higher than the main retrieval threshold to ensure context quality. This conservative threshold may exclude some marginally relevant context sentences that could provide useful narrative continuity.

Chapter Diversity Limit: The maximum of 3 passages per chapter prevents over-representation of theme-rich chapters but may exclude some relevant passages in particularly thematic chapters. This trade-off prioritizes result diversity over exhaustive retrieval.

Expert Review Sample Size: With 2-3 expert reviewers, the qualitative evaluation provides informed feedback but may not capture a full range of interpretive perspectives. Larger expert panels would strengthen validity but exceed thesis resource constraints.

Known-Negative Query Set: The domain validation evaluation uses a curated set of queries known to reference events/characters not in the summaries. While this validates rejection capability, the set may not cover all possible types of invalid queries. Real-world deployment may encounter unanticipated query patterns requiring threshold adjustment.

These limitations and delimitations are acknowledged transparently to contextualize the study's contributions appropriately within the research landscape established in Chapter 2. They also identify clear directions for future work that extends beyond this thesis scope, particularly regarding the scalability of domain-adaptive validation to larger and more diverse literary corpora.

3.12 Summary

This chapter detailed the comprehensive methodology for developing and evaluating a dual-formula hybrid information retrieval system with domain-adaptive semantic validation for thematic analysis of Rizal's novels. The approach integrates:

Theoretical Grounding: Four interconnected frameworks (Domain-Adaptive Pre-training Theory, Hybrid Retrieval Theory, Cross-lingual Transfer Learning Theory, and Thematic Coherence Theory) guide dual-formula system design with semantic grounding validation.

Domain-Adaptive Semantic Validation: A novel validation layer that computes semantic alignment between queries and the corpus domain, preventing retrieval of weakly similar but thematically irrelevant passages. Queries failing domain alignment (< 0.30) and theme proximity (< 0.35) thresholds are explicitly rejected rather than returning hallucinated results.

Preprocessing Pipeline: Filipino-specific stopword handling via stopwords-iso, XLM-RoBERTa tokenization, sentence-level segmentation, stopword-aware weighting, and strict deduplication with corpus vocabulary building for lexical grounding validation.

Dual-Formula Hybrid Architecture:

- **Formula 1 (Main Retrieval):** pairing favoring lexical precision for short queries, semantic understanding for longer queries, with weighted term matching using frequency-based semantic weights
- **Formula 2 (Neighbor Retrieval):** Employs dynamic frequency-based weighting to assess sentence-to-sentence coherence, emphasizing semantic continuity while accounting for lexical similarity through the Jaccard coefficient. This adaptive mechanism prioritizes contextually meaningful and rare words, ensuring smoother thematic transitions between adjacent sentences.

Multi-Component Scoring: Balanced integration of semantic similarity (cosine), weighted lexical overlap, and length normalization in Formula 1; semantic-prioritized scoring with Jaccard-

based lexical similarity in Formula 2 to capture different aspects of document-query proximity within the Vector Space Model framework (Jain et al., 2017).

Thematic Classification: Passage-theme matching using XLM-RoBERTa embeddings for literary analysis beyond simple retrieval.

Comprehensive Evaluation: Quantitative metrics (P/R/F1/MAP) including domain validation metrics (TNR, FPR), ablation studies comparing single-formula vs. dual-formula performance and with/without domain validation, expert review assessing rejection appropriateness, and distribution/co-occurrence analysis.

The methodology directly addresses all research objectives while acknowledging appropriate limitations and delimitations. Implementation details ensure reproducibility, and ethical considerations demonstrate responsible research practices. The dual-formula approach with domain-adaptive validation represents a novel contribution to hybrid retrieval systems by:

1. Recognizing that optimal signal integration depends on retrieval context (query→corpus vs. sentence→sentence)
2. Implementing semantic grounding validation to prevent false retrievals for queries outside the corpus domain
3. Using complementary lexical similarity measures (weighted overlap for query matching, Jaccard for neighbor coherence) appropriate to each retrieval task

The next chapter will present the results obtained through application of this methodology, including quantitative performance metrics, ablation study findings demonstrating the value of

domain validation and dual-formula design, expert evaluation of retrieval quality and rejection decisions, and thematic distribution patterns revealed by the system.

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